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MASTER COVER

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EVOLUTION OF MED-SURG NURSING & ICD CODING

MED-SURG NURSING EVOLUTION: The transition from basic custodial care to evidence-based, highly technical patient-centred practice. **ICD (International Classification of Diseases)** provides standardized disease coding for global epidemiological tracking and clinical health records!

MILESTONE ERA	HISTORICAL & TECHNOLOGICAL EVOLUTION	MODERN CLINICAL NURSING RELEVANCE
1. Custodial Era (Pre-20th Century)	Basic shelter, feeding, and comfort provided by religious orders or untrained aides; zero scientific asepsis.	Highlights the necessity of strict scientific foundations in modern nursing practice.
2. Florence Nightingale Era (Crimean War onwards)	Introduction of environmental sanitation, fresh air, hand hygiene, and systematic patient observation.	Foundational principles of medical and surgical asepsis still govern hospital wards today.
3. Technological Era (Late 20th Century)	Rise of intensive care units, mechanical ventilation, telemetry monitoring, and specialized surgical procedures.	Requires critical thinking, advanced hemodynamic monitoring, and rapid emergency intervention.
4. Evidence-Based Era (Current Standard)	Integration of best research evidence with clinical expertise and patient values (EBP).	Elimination of outdated nursing rituals in favor of proven protocols (e.g., bundle care).

INTERNATIONAL CLASSIFICATION OF DISEASES (ICD-11)

PURPOSE & STRUCTURE

- Maintained by WHO.
- Standardized global classification system for mortality/morbidity.
- Enables seamless health record coding & insurance billing.

NURSE'S ROLE

- Accurate nursing documentation aids medical diagnosis coding.
- Documenting comorbidities and wound staging ensures accurate ICD coding.

EBP TRAP

Performing nursing procedures solely because *"We've always done it this way"* violates EBP standards. Always reference current hospital policy.

🕒 INTRODUCTION SPEED CHECK

Which global organization maintains and updates the ICD disease classification system?

World Health Organization (WHO)

ROLES & RESPONSIBILITIES IN MED-SURG SETTINGS

⚡ **NURSING ROLES:** Med-Surg nurses function across OPDs, inpatient wards, and ICUs. **Core pillars:** Comprehensive Assessment, Nursing Process Execution, Clinical Safety, Patient Advocacy & Interprofessional Coordination!

CLINICAL SETTING	CORE NURSE ROLE & RESPONSIBILITIES	KEY COMPETENCY FOCUS
Outpatient Dept (OPD)	Triage, vital signs screening, health history collection, physician assistance, and discharge instruction education.	Rapid assessment and health promotion.
Inpatient Ward	Comprehensive head-to-toe assessment, medication administration, wound dressing, post-op monitoring, and family liaison.	Execution of the 5-step Nursing Process.
Intensive Care Unit (ICU)	Continuous hemodynamic monitoring, ventilator management, titration of vasoactive drips, and critical crisis intervention.	Advanced life support and physiological stability.

THE 5-STEP NURSING PROCESS IN MEDICAL-SURGICAL CARE

ASSESSMENT & DIAGNOSIS

Gathering data; formulating validated NANDA nursing diagnoses.

PLANNING & ACTION

Setting SMART goals, prioritizing nursing interventions (NIC), executing care.

EVALUATION

Assessing patient response against goals; modifying care plan (NOC).

⚡ **PATIENT ADVOCACY:** The nurse's primary ethical obligation to protect patient safety, uphold informed consent, and question unsafe medical orders.

🎯 **Roles Speed Check**
What is the initial phase of the 5-step nursing process? **Assessment (Data collection and physical examination)**

MEDICAL & SURGICAL ASEPSIS

 **ASEPSIS & INFECTION:** Medical Asepsis ("Clean Technique") reduces pathogenic microorganism numbers; Surgical Asepsis ("Sterile Technique") eliminates ALL microorganisms and spores!

ASEPSIS CATEGORY	OPERATIONAL DEFINITION & SCOPE	CLINICAL NURSING EXEMPLAR
MEDICAL ASEPSIS ("Clean Technique")	Practices designed to reduce total pathogen numbers and prevent spread from person to person.	Hand hygiene, clean gloves, routine environmental cleaning, standard precautions.
SURGICAL ASEPSIS ("Sterile Technique")	Practices eliminating EVERY MICROORGANISM, INCLUDING SPORES , from an object or area.	OR surgeries, Foley catheter insertion, central venous line insertion, sterile dressings.

PRINCIPLES OF SURGICAL ASEPSIS (GOLDEN RULES)

STERILE INTEGRITY RULES

- **Sterile + Sterile = Sterile.** Sterile + Unsterile = Contaminated!
- Only sterile items may be placed within a sterile field.
- Items below waist level are contaminated.
- Moisture passing through sterile barrier contaminates field (Wick effect).

BORDER & AIRFLOW RULES

- **1-Inch Border:** Outer 1 inch of sterile wrapper is considered unsterile.
- Never reach across a sterile field.
- Never talk or sneeze directly over sterile field (wear masks).

WET STERILE FIELD TRAP

If a sterile tray wrapper becomes wet from spilled saline, microorganisms wick through from unsterile table surfaces. Discard immediately!



Asepsis Speed Check

What is the outer border margin of a sterile drape considered?

Unsterile (The outer 1 inch is contaminated border)



 **WOUND HEALING: 3 Intentions:** Primary (clean surgical incision); Secondary (granulation, gaping wound); Tertiary (delayed primary). **4 Stages:** Hemostasis → Inflammatory → Proliferative → Remodeling!

HEALING STAGE	CELLULAR ACTIVITY & TIMELINE	CLINICAL SIGNS & ASSESSMENT
1. HEMOSTASIS	Vasoconstriction, platelet aggregation, and fibrin clot formation.	Active bleeding halts; clot forms.
2. INFLAMMATORY (Days 1–4)	Vasodilation, neutrophils and macrophages phagocytose debris and bacteria.	Erythema, warmth, edema, pain (Cardinal signs).
3. PROLIFERATIVE (Days 4–21)	Fibroblasts deposit collagen; angiogenesis; granulation tissue formation.	Pink/red beefy Granulation Tissue .
4. REMODELING (Day 21–2Yr)	Collagen cross-linking and strengthening; scar tissue remodeling.	Pale, flat, mature scar tissue forms.

FACTORS DELAYING HEALING & COMPLICATIONS

SYSTEMIC DELAYS

- • **Diabetes Mellitus:** Impaired microvascular perfusion.
- • **Malnutrition:** Protein, Vit C, Zinc deficiencies.
- • **Steroids:** Suppresses inflammation.

COMPLICATIONS

- • **Dehiscence:** Partial/total rupture of wound layers.
- • **Evisceration:** Protrusion of viscera (**Cover with sterile saline dressings!**).



Healing Speed Check

Emergency nursing action for wound evisceration?

Cover protruding viscera immediately with sterile saline-soaked dressings



WOUND CARE & STERILE DRESSING TECHNIQUE

⚡ **WOUND DRESSING:** Sterile technique prevents surgical site infections (SSIs). **Golden Rule:** Clean from cleanest area (wound center) outward! **Exudate:** Serous, Sanguineous, Serosanguineous, Purulent.

DRESSING STEP	PROCEDURAL PROTOCOL & RATIONALE	CRITICAL NURSING PRECAUTION
1. PREPARATION	Review orders, gather supplies, medicate patient for pain 30 mins prior.	Premedicate to control dressing change discomfort.
2. REMOVAL	Don clean gloves, remove old soiled dressing gently (use saline if stuck).	Never pull forcefully; dispose of biohazardous waste.
3. CLEANSING	Don sterile gloves. Clean wound from INCISION LINE OUTWARD using separate sterile swabs.	Never pass used swab back over cleansed tissue!
4. APPLICATION	Apply sterile contact layer, gauze pads, and secure without restricting circulation.	Document wound dimensions and exudate characteristics.

WOUND DRAINAGE EXUDATE TYPES

EXUDATE TYPE	APPEARANCE & COMPOSITION	CLINICAL SIGNIFICANCE
Serous	Clear, watery, straw-colored plasma.	Normal clean healing.
Sanguineous	Bright red, bloody drainage.	Active bleeding.
Purulent	Thick, opaque, yellow/green with foul odor.	Bacterial infection / pus!



Wound Care Speed Check

Correct cleansing direction during sterile wound dressing change?

From incision line outward toward surrounding skin



 **PRE-OP CARE:** Comprehensive preparation prior to surgery. **Informed Consent:** Surgeon informs, Nurse witnesses signature! **NPO status** prevents aspiration.

PRE-OP DOMAIN	NURSING ACTIONS & RESPONSIBILITIES	NORCET HIGH-YIELD RATIONALE
LEGAL CONSENT	Verify informed consent is signed and attached BEFORE administering pre-op sedatives!	Surgeon informs; Nurse witnesses. Invalid if signed under sedation.
NPO STATUS	Enforce fasting (Clear liquids 2 hrs, breast milk 4 hrs, light meal 6 hrs, fatty meal 8 hrs).	Prevents aspiration of gastric contents during intubation.
INVESTIGATIONS	Review baseline vitals, CBC, coagulation, electrolytes, ECG, and chest X-ray.	Detects underlying abnormalities and establishes intra-op baseline.
PRE-OP PREP	Clip surgical site (no razors!), remove dentures, jewelry, nail polish, and contact lenses.	Reduces bacterial load; prevents airway obstruction and pulse oximeter interference.

INFORMED CONSENT SEDATION TRAP

A pre-op patient has received Midazolam. Getting consent signed now is **illegal and invalid**. Surgeon must wait until patient is fully alert.


Pre-Op Speed Check
 Nurse's legal role regarding surgical informed consent?

Witness signature and verify consent form is present on chart

POST-OPERATIVE NURSING CARE & COMPLICATIONS

 **POST-OP CARE:** PACU transition focusing on **ABCDE**. Prevention of pulmonary atelectasis, DVT, and surgical hemorrhage is paramount!

POST-OP PHASE	NURSING ASSESSMENT & INTERVENTIONS	COMPLICATION PREVENTED
PACU TRANSITION	Maintain patent airway, monitor SpO ₂ , vitals q15m, inspect surgical drains/dressings.	Airway obstruction, hypovolemic shock.
RESPIRATORY HYGIENE	Encourage deep breathing and coughing (DBCE) hourly, use incentive spirometer.	Atelectasis & Pneumonia.
DVT PROPHYLAXIS	Apply SCDs / TED stockings, early leg exercises, subcutaneous LMWH.	Deep Vein Thrombosis & Pulmonary Embolism (PE).
GI / URINARY	Assess bowel sounds, passing of flatus, monitor urine output (> 0.5 mL/kg/hr).	Paralytic ileus, urinary retention.

HEMORRHAGE SIGNS

- Tachycardia, falling BP, cool pale skin, restlessness.
- Increasing drain output. Apply direct pressure!

INCENTIVE SPIROMETRY

- Inhale maximally, hold 3–5 seconds, exhale slowly.
- 10 repetitions every hour while awake.

 **Post-Op Speed Check**
Earliest vital sign indicator of internal post-operative hemorrhage? **Tachycardia (Heart rate elevation precedes BP drop)**

ALTERNATIVE THERAPIES IN MED-SURG CARE

⚡ **ALTERNATIVE THERAPIES (CAM):** Integration of non-conventional therapies. **Herbal supplements, Yoga, Acupuncture, Meditation.** **Safety and drug interactions are primary nursing concerns!**

CAM MODALITY	THERAPEUTIC PRINCIPLE & PRACTICE	CLINICAL NURSING PRECAUTION
HERBAL SUPPLEMENTS	Plant remedies (Ginkgo, Garlic, Ginseng, St. John's Wort) used for chronic ailments.	Severe bleeding risk! Herbs inhibit platelet aggregation and interact with anticoagulants (Warfarin).
MIND-BODY PRACTICES	Yoga, meditation, progressive muscle relaxation, guided imagery.	Safe adjuncts for chronic pain, anxiety reduction, and hypertension control.
ACUPUNCTURE	Insertion of sterile needles into body meridians to release endorphins.	Assess for bleeding disorders or skin infection prior to referral.

PRE-OP HERBAL STOP TRAP

Patients scheduled for surgery must stop all herbal supplements (Garlic, Ginkgo, Ginseng, Vitamin E) **at least 1 to 2 weeks prior to surgery** to prevent hemorrhage!



Alternative Therapies Speed Check

How many weeks prior to surgery must herbal anticoagulants like Ginkgo be stopped? **At least 1 to 2 weeks prior to surgery**



OT ORGANIZATION, DESIGN & ZONING

 **OT ZONING:** Environmental traffic control prevents surgical site infections (SSIs). **4 Zones:** Unrestricted (Street clothes), Semi-Restricted (Scrubs/caps), Restricted (Masks mandatory), Sterile Core!

OT ZONE	ACCESS RULES & ATTIRE REQUIREMENT	OPERATIONAL AREA & FUNCTION
1. UNRESTRICTED	Street clothes permitted; transition area for patients and administrative admission.	Holding area, reception desk, locker rooms, offices.
2. SEMI-RESTRICTED	Hospital scrub suits, caps, and shoe covers required. No street clothes.	Corridors, instrument processing, sterile storage.
3. RESTRICTED	Scrubs + Caps + Face Masks mandatory at all times!	Operating Rooms (ORs), scrub sink areas, sterile core.

OT ENVIRONMENTAL CONTROLS

POSITIVE AIR PRESSURE

- ORs maintain **Positive Air Pressure** relative to corridors.
- Air flows *outward* when doors open, preventing dust and bacteria from entering.

CLIMATE CONTROL

- Temperature: **20°C to 24°C (68°F–75°F)**.
- Humidity: 30%–60% to prevent static electricity and bacterial proliferation.



OT Setup Speed Check

What air pressure system is maintained inside operating theatres?

Positive Pressure Airflow



SCRUB NURSE VS CIRCULATING NURSE ROLES

 **SURGICAL TEAM ROLES:** **Scrub Nurse is STERILE** (handles instruments, assists surgeon directly); **Circulating Nurse is NON-STERILE** (manages room, documents, records surgical counts)!

SURGICAL ROLE	STERILE STATUS	CORE RESPONSIBILITIES & DUTIES	KEY COLLABORATION
SCRUB NURSE	STERILE (Gown & gloves)	• Sets up sterile instrument table. • Passes instruments to surgeon. • Participates in sponge/instrument counts.	Works directly opposite surgeon in sterile field.
CIRCULATING NURSE	NON-STERILE (Scrubs & cap)	• Manages unsterile OT environment. • Patient positioning & skin prep. • Maintains documentation & specimen labeling.	Coordinates with anesthesia, pathology, and central supply.

SURGICAL COUNT PROTOCOL

- Sponges, needles, and instruments counted **3 times**: 1) Baseline; 2) Cavity closure; 3) Skin closure.
- Incorrect count halts surgery!

 **SPECIMEN RULE:** Circulating nurse is responsible for correct labeling and preservation of all biopsy specimens. Never place in saline unless ordered!



Scrub vs Circulating Speed Check

Which surgical team member is unsterile and manages counts?

The Circulating Nurse



SURGICAL POSITIONS & NERVE PROTECTION

⚡ **SURGICAL POSITIONS:** Patient positioning must ensure exposure while preventing nerve palsies and pressure ulcers. **Supine, Prone, Lithotomy, Trendelenburg, Lateral!**

POSITION	ALIGNMENT & INDICATIONS	VULNERABLE NERVES / PRESSURE POINTS	PHYSIOLOGICAL RISK
SUPINE	Flat on back (cardiac, abdominal, hernia).	Occiput, elbows, sacrum, heels.	Pressure sores.
TRENDELENBURG	Head tilted down 15°–30° (pelvic/gynecological).	Brachial plexus compression.	Increased venous return; reduced lung excursion.
LITHOTOMY	Legs elevated in stirrups (urological, perineal, ob-gyn).	Common peroneal nerve (compression = footdrop!).	Orthostatic hypotension upon lowering legs.
PRONE	Face down on chest bolsters (spinal, back).	Eyes (corneal abrasion), ulnar nerve.	IVC compression, respiratory compromise.

LITHOTOMY FOOTDROP TRAP

Improper stirrup padding compressing the fibular head damages the **Common Peroneal Nerve**, resulting in post-op **Footdrop** (inability to dorsiflex foot).

**Positions Speed Check**

Which nerve is compressed during improper lithotomy positioning causing footdrop? **Common Peroneal (Fibular) Nerve**



SURGICAL INSTRUMENTS, SUTURES & MATERIALS

⚡ **INSTRUMENTS & SUTURES: 4 families: Cutting (scalpels, scissors), Grasping (forceps), Hemostatic (clamps), Retracting (Senn, Richardson). Sutures:**
Absorbable (Vicryl) vs Non-Absorbable (Nylon)!

INSTRUMENT FAMILY	FUNCTION & MECHANISM	CLINICAL EXAMPLES
1. CUTTING	Sharp blades used for incising skin and fascia.	Scalpels (#10, #15), Mayo & Metzenbaum scissors.
2. HEMOSTATIC	Ratcheted locking instruments to clamp bleeding vessels.	Mosquito, Crile, and Kelly clamps.
3. GRASPING	Thumb forceps and ring clamps for holding tissues/drapes.	Adson forceps, Allis clamps, Babcock clamps.
4. RETRACTING	Blades used to pull back wound edges for exposure.	Senn, Richardson, Deaver, and Gelpi retractors.

SURGICAL SUTURES

ABSORBABLE SUTURES

- Degraded by body enzymes over weeks/months.
- **Vicryl**: Synthetic braided (60–90 days).
- **Plain/Chromic Catgut**: Natural mucosal use.

NON-ABSORBABLE SUTURES

- Permanent tensile strength; not digested by body.
- **Silk**: Braided, high knot security.
- **Nylon / Prolene**: Synthetic monofilament for skin.



Instruments Speed Check

Primary function of a hemostatic clamp (e.g., Kelly clamp)?

To clamp bleeding blood vessels and achieve hemostasis



DISINFECTION & STERILIZATION (SPALDING CLASSIFICATION)

⚡ **SPALDING CLASSIFICATION:** Critical (sterile tissue → Sterilization); Semi-Critical (mucous membranes → High-Level Disinfection); Non-Critical (intact skin → Low Disinfection). Autoclave: 121°C at 15 psi!

SPALDING TIER	PATIENT CONTACT POINT	REQUIRED PROCESSING LEVEL & AGENT	EXEMPLAR
CRITICAL	Enter sterile tissues or vascular system. High risk.	STERILIZATION (Steam autoclave, ETO, Hydrogen Peroxide plasma).	Surgical instruments, implants.
SEMI-CRITICAL	Touch mucous membranes or non-intact skin. Moderate risk.	HIGH-LEVEL DISINFECTION (Glutaraldehyde 2%, OPA).	Endoscopes, endotracheal tubes.
NON-CRITICAL	Touch intact skin only. Low risk.	LOW-LEVEL DISINFECTION (Quaternary ammonium, alcohol).	Stethoscopes, BP cuffs, bedpans.

AUTOCLAVE & BIOLOGICAL INDICATORS**AUTOCLAVE PARAMETERS**

- • 121°C (250°F) at 15 psi for 15–30 mins. Kills all spores.

BIOLOGICAL INDICATOR

- • *Geobacillus stearothermophilus* spore test (gold standard proving sterility!).

**Sterilization Speed Check**

Biological spore strip used to test steam autoclaves?

Geobacillus stearothermophilus



SURGICAL SCRUBBING, GOWNING & GLOVING

 **SURGICAL SCRUB:** Duration 3–5 minutes. **Elbows always kept HIGHER than hands** so water drips off elbows. Closed-glove technique prevents skin contact with outer glove!

SCRUB STEP	PROCEDURAL PROTOCOL & ANATOMICAL RULE	CRITICAL PRECAUTION
1. PREPARATION	Remove rings, watches. Inspect hands for cuts. Put on surgical cap and mask.	Masks must cover nose and mouth completely.
2. HAND SCRUB	Scrub fingers, nails, hands, and forearms up to 2 inches above the elbow .	Keep hands HIGHER than elbows so water drips off elbows, preventing backflow!
3. DRYING & GOWNING	Dry arms systematically with sterile towel. Don sterile gown without touching outer surface.	Perform Closed-Glove Technique inside sleeve cuffs.

CONTAMINATION RULE

If a sterile glove or gown touches an unsterile surface during surgery, it is immediately contaminated! Un-glove and re-glove instantly.



Scrubbing Speed Check

Why must hands be kept higher than elbows during surgical scrubbing?

To ensure water and contaminants drip downward off elbows



INTRAOPERATIVE MONITORING & WHO SAFETY CHECKLIST

 **WHO SURGICAL CHECKLIST:** Mandatory safety protocol. **3 Phases: Sign In (Before induction) → Time Out (Before incision) → Sign Out (Before leaving OR)!**
Capnography is gold standard for ETT placement.

CHECKLIST PHASE	TIMING EXECUTION	CORE VERIFICATION ITEMS	TEAM
1. SIGN IN	Before anesthesia induction.	Confirm identity, site marked, consent, allergy check.	Nurse & Anesthesia
2. TIME OUT	Before skin incision.	Team introductions, re-verify name, procedure, site, antibiotic given.	Entire Surgical Team
3. SIGN OUT	Before leaving OR.	Confirm instrument/sponge counts, specimen labeling review.	Nurse, Surgeon, Anesthesia

MONITORING & HAZARDS

MONITORING STANDARDS

- ECG, SpO2, NIBP q5m, Core temperature.
- **Capnography (EtCO2):** ETT placement confirmation.

ELECTROCAUTERY SAFETY

- Place grounding dispersive pad on large muscle mass close to surgical site, away from metal implants.



Safety Speed Check

Gold standard monitor verifying correct endotracheal tube placement?

Capnography (End-Tidal CO2 / EtCO2)



ANAESTHESIA TYPES & MALIGNANT HYPERTHERMIA

 **ANAESTHESIA & MH:** General, Spinal, Epidural, Local. **Malignant Hyperthermia (MH)** is a hypermetabolic muscle crisis triggered by volatile agents/succinylcholine. **Antidote: Dantrolene Sodium IV!**

ANAESTHESIA	MECHANISM & ROUTE	INDICATIONS	MAJOR COMPLICATIONS
GENERAL	Reversible CNS depression via IV/inhalation. ETT required.	Major surgeries, prolonged operations.	Aspiration, Malignant Hyperthermia, PONV.
SPINAL	Local anesthetic injected into subarachnoid CSF at L3–L4. Awake patient.	Lower abdominal, C-section, urological.	PDPH (Headache) , severe hypotension.
EPIDURAL	Local anesthetic injected into epidural space outside dura mater.	Labor analgesia, post-op pain control.	Accidental dural puncture, hypotension.

MALIGNANT HYPERTHERMIA CRISIS

- Early signs: Tachycardia, rising EtCO₂, **masseter muscle rigidity**.
- **Emergency Action:** Stop trigger gas, give 100% O₂, administer **Dantrolene Sodium IV**, and active cooling!



Anaesthesia Speed Check

Specific pharmacological antidote for Malignant Hyperthermia crisis?

Dantrolene Sodium IV

FLUID & ELECTROLYTE IMBALANCES MASTER

ELECTROLYTES MASTER: Potassium (K^{+}) Normal: 3.5–5.0 mEq/L) governs cardiac rhythm (**Hyperkalemia causes peaked T waves & VFib**). Sodium (Na^{+}) governs neurological function. Trousseau's & Chvostek's signs test Hypocalcemia!

ELECTROLYTE	NORMAL RANGE	CLINICAL IMBALANCE & MANIFESTATIONS	CRITICAL NURSING & ECG ACTION
POTASSIUM (K^{+})	3.5–5.0 mEq/L	Hypo: Weakness, flattened T waves, U waves. Hyper: PEAKED T WAVES, widened QRS, VFib!	- Never push IV KCl undiluted! - Treat high K^{+} with Insulin + Dextrose, Calcium Gluconate.
SODIUM (Na^{+})	135–145 mEq/L	Hypo: Headache, lethargy, confusion, seizures. Hyper: Thirst, dry mucous membranes, agitation.	Restrict free water in hyponatremia; monitor GCS closely.
CALCIUM (Ca^{2+})	8.5–10.5 mg/dL	Hypo: Tetany, spasms, positive Trousseau's & Chvostek's signs.	Assess for carpopedal spasm and laryngeal stridor post-thyroidectomy.

IV POTASSIUM PUSH TRAP

Pushing IV Potassium Chloride bolus causes instant cardiac arrest! Dilute in IV fluids, infuse at max 10 mEq/hr, and ensure urine output.



Electrolytes Speed Check

ECG abnormality classically associated with severe hyperkalemia?

Peaked T waves followed by widened QRS complexes



SHOCK CLINICAL MANAGEMENT & EMERGENCY PROTOCOLS

SHOCK CLASSIFICATION: Systemic cellular hypoperfusion leading to anaerobic metabolism and lactic acidosis. **4 Types: Hypovolemic, Cardiogenic, Distributive (Septic/Anaphylactic), Obstructive!** Tachycardia is the first sign of shock!

SHOCK TYPE	PATHOPHYSIOLOGICAL DEFECT	CLINICAL PRESENTATION & HEMODYNAMICS	PRIMARY NURSING INTERVENTION
HYPOVOLEMIC	Massive intravascular fluid loss (hemorrhage, dehydration).	Tachycardia, hypotension, cool clammy skin, flat neck veins, oliguria.	Rapid IV fluid resuscitation (Crystalloids/Blood), stop bleeding.
CARDIOGENIC	Primary pump failure (Massive myocardial infarction).	Tachycardia, hypotension, pulmonary crackles, JVD, S3 heart sound.	Inotropic support (Dobutamine). Avoid large fluid boluses!
DISTRIBUTIVE	Severe peripheral vasodilation (Septic / Anaphylactic).	Warm flushed skin initially (septic), fever, urticaria, bronchospasm.	Antibiotics + fluids + Norepinephrine (septic); Epinephrine (anaphylactic).
OBSTRUCTIVE	Mechanical obstruction to blood flow (Tamponade, Tension Pneumothorax).	Hypotension, muffled heart sounds, tracheal deviation, absent breath sounds.	Emergency decompression (Pericardiocentesis / Needle thoracostomy).

CARDIOGENIC SHOCK FLUID TRAP

Administering rapid IV fluid boluses in *Cardiogenic Shock* worsens pulmonary congestion and causes cardiac arrest!

**Shock Speed Check**

Earliest physiological vital sign indicator of impending shock?

Tachycardia (Elevated heart rate compensating for dropping stroke volume)



RESPIRATORY ASSESSMENT & DIAGNOSTIC STUDIES

 **RESPIRATORY ASSESSMENT:** Evaluation of upper and lower respiratory tracts. **Key physical findings:** Cyanosis (central vs peripheral), clubbing, barrel chest, adventitious breath sounds (crackles, wheezes, rhonchi, pleural friction rub)!

ASSESSMENT METHOD	KEY INSPECTION, PALPATION, PERCUSSION & AUSCULTATION FINDINGS	CLINICAL NURSING SIGNIFICANCE
INSPECTION	Respiratory rate/depth, use of accessory muscles, chest symmetry, anteroposterior-to-transverse diameter ratio (barrel chest in COPD).	Detects signs of respiratory distress and chronic air trapping.
PALPATION & PERCUSSION	Tactile fremitus (vibrations); percussion resonance (hyperresonance in pneumothorax/emphysema; dullness in effusion/consolidation).	Identifies underlying density or air-filled pleural spaces.
AUSCULTATION	Normal vesicular/bronchovesicular sounds vs. adventitious sounds: Crackles (fluid), Wheezes (bronchospasm), Rhonchi (mucus), Friction rub.	Pinpoints alveolar fluid, airway narrowing, or pleural inflammation.

DIAGNOSTIC STUDIES IN RESPIRATORY DISORDERS

PFT & BRONCHOSCOPY

- • **Pulmonary Function Tests (PFT):** Measures lung volumes, FEV1, FVC (diagnoses obstructive vs restrictive disease).
- • **Bronchoscopy:** Direct visualization of tracheobronchial tree. *Post-op rule: Keep NPO until gag reflex returns!*

THORACENTESIS & SPUTUM

- • **Thoracentesis:** Aspiration of pleural fluid. *Position client upright leaning over bedside table.* Max 1000–1500 mL removed at once.
- • **Sputum Culture:** Obtain early morning sterile specimen prior to antibiotics.

POST-BRONCHOSCOPY GAG REFLEX TRAP

Offering food or water immediately following diagnostic bronchoscopy before the local anesthetic wears off causes **massive aspiration and asphyxiation**. Always verify gag reflex first!



Respiratory Assessment Speed Check

What is the critical post-procedure nursing priority following bronchoscopy?

Keep client NPO until gag reflex fully returns to prevent aspiration

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NUR-08 Batch 2: Units IV, V & VI
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CHRONIC OBSTRUCTIVE PULMONARY DISEASE (COPD)

 **COPD PATHOPHYSIOLOGY:** Progressive airflow limitation caused by chronic bronchitis (airway inflammation and excessive mucus) and emphysema (destruction of alveolar walls and loss of elasticity). **Hypoxic drive is the primary breathing stimulus!**

COPD PHENOTYPE	CLINICAL NICKNAME & PATHOLOGICAL HALLMARKS	KEY PHYSICAL PRESENTATION
CHRONIC BRONCHITIS	"Blue Bloaters" Chronic airway inflammation, hypertrophy of mucous glands, severe productive cough, and cyanosis.	Cyanotic, peripheral edema (cor pulmonale), polycythemia, wheezing, and coarse crackles.
EMPHYSEMA	"Pink Puffers" Destruction of alveolar air sacs, loss of elastic recoil, air trapping, and enlarged alveolar dead space.	Barrel chest, pursed-lip breathing, use of accessory muscles, thin cachectic appearance, pink skin tone.

OXYGEN THERAPY TRAP & NURSING MANAGEMENT IN COPD

THE HYPOXIC DRIVE TRAP

- In advanced COPD, chronic hypercapnia (PaCO_2 elevation) desensitizes the brain's respiratory center. Breathing is driven entirely by **hypoxia** (low PaO_2).
- **FATAL ERROR:** Administering high-flow oxygen suppresses the hypoxic drive, causing apnea and respiratory arrest!
- **✓ Rule:** Titrate low-flow oxygen via nasal cannula to maintain **SpO₂ 88%–92%**.

COPD NURSING INTERVENTIONS

- **Pursed-Lip Breathing:** Prolongs exhalation, prevents airway collapse, and promotes CO₂ elimination.
- **Diaphragmatic Breathing:** Strengthens diaphragm.
- High-calorie, nutrient-dense, small frequent meals (reduces diaphragmatic crowding).
- Chest physiotherapy and postural drainage.



COPD Speed Check

What is the target SpO₂ saturation range when administering oxygen to a COPD client? **88% to 92% (Avoid high-flow O₂ to prevent suppressing hypoxic drive)**

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BRONCHIAL ASTHMA & STATUS ASTHMATICUS

 **BRONCHIAL ASTHMA:** Chronic inflammatory airway disorder characterized by reversible airway obstruction, bronchospasm, mucosal edema, and excessive thick mucus production. **Status asthmaticus is a life-threatening medical emergency unresponsive to standard bronchodilators!**

CLINICAL PHASE	PATHOPHYSIOLOGICAL MANIFESTATIONS & TRIGGERS	EMERGENCY NURSING MANAGEMENT
ACUTE ASTHMA ATTACK	Triggered by allergens (pollen, dust mites, smoke), exercise, cold air, or emotional stress. Exhibits expiratory wheezing, chest tightness, and dyspnea.	Administer short-acting beta-2 agonist (SABA / Albuterol) via nebulizer or MDI. Sit patient in high-Fowler's position.
STATUS ASTHMATICUS	Severe, prolonged, unresponsive bronchospasm with hyperinflation, tachypnea, and cyanosis. May progress to respiratory failure and death.	EMERGENCY INTUBATION PREPARATION Prepare for intubation if patient is unresponsive to medical therapy. Administer SABA and steroids. Monitor vital signs and oxygen saturation.

PEAK EXPIRATORY FLOW RATE (PEFR) ZONES & INHALER EDUCATION

PEAK FLOW METER ZONES

- **Green Zone (80%–100% personal best):** Asthma is well controlled.
- **Yellow Zone (50%–80%):** Caution; acute flare-up; use rescue inhaler.
- **Red Zone (<50%):** MEDICAL EMERGENCY! Immediate SABA use + seek emergency care.

MDI WITH SPACER INSTRUCTIONS

1. Shake inhaler, attach to spacer.
2. Exhale completely, place mouthpiece between teeth and seal lips.
3. Depress canister once, take slow deep breath over 5 seconds.
4. Hold breath for 10 seconds, then exhale slowly. *Rinse mouth after steroid inhalers (prevents oral candidiasis/thrush!)*



Asthma Speed Check

What is the most critical nursing action following administration of inhaled corticosteroids?

Instruct patient to rinse mouth thoroughly with water to prevent oral thrush

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⚡ **PLEURAL & PULMONARY INFECTIONS:** Accumulation of fluid in pleural space (Pleural Effusion) or pus (Empyema). **Pneumonia and Bronchiectasis** impair gas exchange and alveolar ventilation!

CONDITION	PATHOPHYSIOLOGICAL ETIOLOGY	KEY CLINICAL FINDINGS	PRIMARY NURSING INTERVENTION
PLEURAL EFFUSION	Excess fluid accumulation in pleural space (Transudate from HF; Exudate from infection/malignancy).	Decreased breath sounds, dull percussion note, flat tactile fremitus, dyspnea.	Assist with Thoracentesis ; monitor for pneumothorax post-procedure.
EMPYEMA	Accumulation of thick, purulent exudate (pus) within the pleural cavity, secondary to bacterial pneumonia.	High-grade fever, chills, pleural chest pain, sepsis, debilitation.	Chest tube insertion for drainage, IV antibiotic therapy, thoracotomy.
PNEUMONIA	Inflammatory infection of lung parenchyma (alveoli fill with exudate and WBCs). Bacterial, viral, or aspiration.	Fever, productive cough with rust-colored sputum (pneumococcal), pleuritic pain, crackles.	Oxygen therapy, sputum cultures, IV antibiotics, incentive spirometry, hydration.

THORACENTESIS FLUID VOLUME TRAP

During thoracentesis, **never remove more than 1,000 to 1,500 mL of pleural fluid at one time!**

✓ **RATIONALE:** Rapid removal alters intrapleural pressure too quickly, precipitating *re-expansion pulmonary edema and cardiovascular collapse!*

⚡ **PNEUMOCOCCAL SPUTUM:**

Classic textbook sign of pneumococcal pneumonia is the production of **rust-colored (blood-tinged) sputum**, accompanied by abrupt high fever and shaking chills.

 **Pleural & Pulmonary Speed Check**
Maximum volume of pleural fluid safely removed during a single thoracentesis? **1,000 to 1,500 mL (Prevents re-expansion pulmonary edema)**

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CHEST INJURIES, PNEUMOTHORAX & FLAIL CHEST

CHEST TRAUMA: Blunt or penetrating chest injuries disrupt negative intrapleural pressure. **Tension Pneumothorax is a fatal medical emergency causing mediastinal shift and obstructive shock!**

CHEST TRAUMA TYPE	PATHOPHYSIOLOGICAL MECHANISM	CLINICAL MANIFESTATIONS	EMERGENCY ACTION
TENSION PNEUMOTHORAX	Air enters pleural space on inspiration but cannot escape (one-way valve). Trachea shifts to HEALTHY SIDE!	Severe respiratory distress, absent breath sounds, tracheal deviation, jugular venous distension (JVD), hemodynamic collapse.	EMERGENCY NEEDLE DECOMPRESSION (2nd ICS, mid-clavicular line) + Chest tube insertion!
OPEN PNEUMOTHORAX ("Sucking Chest Wound")	Penetrating chest wall defect allows atmospheric air to flow freely in and out of pleural space.	Sucking sound on inspiration, bubbling wound, respiratory distress.	Apply sterile three-sided occlusive dressing (allows air to escape, prevents entry).
FLAIL CHEST	Fracture of 3 or more adjacent ribs in 2 or more places, causing an independent unstable chest wall segment.	Paradoxical chest movement (segment moves inward on inspiration, outward on expiration), severe pain.	High-flow O ₂ , pain control, mechanical ventilation stabilization.

SUCKING CHEST WOUND 4-SIDED TAPE TRAP

✗ **FATAL ERROR:** Taping an occlusive dressing on all 4 sides over an open pneumothorax.
✓ **CORRECT ACTION:** Tape on **3 sides only!** This creates a flutter-valve effect: air escapes during exhalation but cannot enter during inhalation, preventing tension pneumothorax!

⚡ TRACHEAL DEVIATION RULE:

In Tension Pneumothorax, the trachea deviates **AWAY FROM THE AFFECTED SIDE** (toward the healthy uninjured lung) due to massive positive pressure pushing mediastinal structures!

**Chest Trauma Speed Check**

Where is emergency needle decompression performed for tension pneumothorax?

2nd Intercostal Space, Mid-Clavicular Line on affected side

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 **CHEST TUBE MANAGEMENT:** Restores negative intrapleural pressure. 3-Chamber System: Collection, Water-Seal, and Suction Control. **Continuous bubbling in water-seal chamber indicates an air leak; Tiding is normal!**

CHAMBER / PARAMETER	FUNCTIONAL MECHANISM & NORMAL FINDING	ABNORMAL FINDING & NURSING ACTION
1. COLLECTION CHAMBER	Collects blood and pleural fluid draining from chest cavity. Measure output q1h initially.	Output > 100 mL/hr or bright red blood indicates active hemorrhage! Notify surgeon.
2. WATER-SEAL CHAMBER	Acts as one-way valve allowing air to exit while preventing backflow. Tidaling (water level rises on inspiration, falls on expiration) is normal!	• Cessation of tidaling: Lung re-expanded OR tubing kinked/obstructed. • Continuous bubbling: Air leak in system!
3. SUCTION CONTROL	Regulates amount of negative suction pressure applied (usually set to -20 cmH2O via gentle bubbling).	Ensure gentle continuous bubbling in suction chamber (vigorous bubbling evaporates water too fast).

EMERGENCY BEDSIDE PROTOCOLS FOR ACCIDENTS

ACCIDENTAL DISCONNECTION

- If chest tube disconnects from drainage system:
- **✓ IMMEDIATE ACTION:** Submerge the distal end of the chest tube in a **bottle of sterile water** (creates temporary water seal) OR clamp momentarily with rubber-tipped hemostat (briefly only).

ACCIDENTAL TUBE PULL-OUT

- If chest tube pulls entirely out of chest wall:
- **✓ IMMEDIATE ACTION:** Cover wound immediately with a **sterile petroleum jelly (vaseline) gauze dressing taped on 3 sides**, and notify physician stat!



Chest Tube Speed Check

What does continuous bubbling in the water-seal chamber indicate?

An air leak in the chest tube system or patient pleural space

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ARDS & ACUTE RESPIRATORY FAILURE

ARDS (ACUTE RESPIRATORY DISTRESS SYNDROME): Severe non-cardiogenic pulmonary edema characterized by bilateral infiltrates on CXR and refractory hypoxemia ($\text{PaO}_2/\text{FiO}_2$ ratio ≤ 200). **Requires mechanical ventilation with PEEP!**

CLINICAL STAGE	PATHOPHYSIOLOGICAL MECHANISM	CLINICAL PRESENTATION & MANAGEMENT
1. INJURY / EXUDATIVE (First 24–72 hrs)	Damage to alveolar-capillary membrane by sepsis, aspiration, or trauma $\rightarrow$ Protein-rich fluid floods alveoli $\rightarrow$ Surfactant inactivation & atelectasis.	Dyspnea, tachypnea, refractory hypoxemia (unresponsive to high-flow O ₂ mask). CXR shows diffuse bilateral "white out".
2. FIBROPROLIFERATIVE (Days 7–14)	Inflammatory destruction leads to pulmonary fibrosis, alveolar scarring, and decreased lung compliance.	Worsening hypoxemia, increased work of breathing, pulmonary hypertension.
3. RESOLUTION (After 21 days)	Lung tissue heals over weeks to months, though chronic restrictive deficits may persist.	Gradual weaning from mechanical ventilation, physical rehabilitation.

MECHANICAL VENTILATION & PRONE POSITIONING IN ARDS

ROLE OF PEEP (POSITIVE END-EXPIRATORY PRESSURE)

- Applies positive pressure at end of exhalation to keep collapsed alveoli open and improve oxygenation.
- **Complication:** High PEEP increases intrathoracic pressure, decreasing venous return $\rightarrow$ **Hypotension & decreased cardiac output!**

PRONE POSITIONING PROTOCOL

- Placing severe ARDS patients face-down (prone) for 16 hours/day.
- **Mechanism:** Redistributes blood flow to dorsal lung regions, recruiting collapsed alveoli and dramatically improving $\text{PaO}_2/\text{FiO}_2$ ratio!



ARDS Speed Check

What is the primary hemodynamic complication of high PEEP settings on mechanical ventilators?

Hypotension and decreased cardiac output due to increased intrathoracic pressure

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PULMONARY EMBOLISM & OXYGEN DELIVERY DEVICES

 **PULMONARY EMBOLISM (PE):** Obstruction of pulmonary arterial bed by a thrombus (usually originating from deep veins of lower extremities / DVT).
Classic Triad: Sudden dyspnea, pleuritic chest pain, and unexplained tachypnea! Medical emergency.

DIAGNOSTIC / TREATMENT	CLINICAL PROTOCOL & RATIONALE	NURSING PRECAUTION
DIAGNOSTIC WORKUP	D-Dimer blood test (elevated clotting breakdown), Spiral CT Pulmonary Angiography (CTPA / gold standard), V/Q scan.	Assess patient for IV contrast allergy (iodine/ shellfish) prior to CTPA.
ANTICOAGULATION & LYDIC	Immediate IV Unfractionated Heparin or LMWH bridging to oral anticoagulants. Massive PE with shock requires Thrombolytic therapy (rt-PA).	Monitor aPTT (for Heparin, target 1.5–2.5x control) and INR (for Warfarin, target 2.0–3.0). Keep Protamine Sulfate at bedside!

OXYGEN DELIVERY DEVICES & FIO2 PRECISION

LOW-FLOW OXYGEN DEVICES

- **Nasal Cannula:** 1–6 L/min delivers 24%–44% FiO₂. Comfortable, allows eating/talking.
- **Simple Face Mask:** 5–8 L/min delivers 40%–60% FiO₂ (min flow 5 L/min to flush CO₂).
- **Non-Rebreather Mask (NRB):** 10–15 L/min delivers 80%–100% FiO₂ (reservoir bag must remain inflated!).

HIGH-FLOW OXYGEN DEVICES

- **Venturi Mask:** Delivers precise, controlled FiO₂ (24%–50%) via interchangeable color-coded jet adapters. Ideal for COPD clients!
- **High-Flow Nasal Cannula (HFNC):** Delivers heated, humidified O₂ up to 60 L/min.



PE & Oxygen Speed Check

What is the specific antidote for acute Heparin overdose?

Protamine Sulfate IV

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ABG INTERPRETATION & MECHANICAL VENTILATION

ABG INTERPRETATION (ROME RULE): Arterial Blood Gases assess acid-base balance. **ROME: Respiratory Opposite** (pH and PaCO₂ move in opposite directions); **Metabolic Equal** (pH and HCO₃ move in same direction)! Normal pH: 7.35–7.45; PaCO₂: 35–45 mmHg; HCO₃: 22–26 mEq/L.

ACID-BASE DISORDER	PRIMARY BLOOD GAS ABNORMALITY	COMMON CLINICAL ETIOLOGY	COMPENSATORY MECHANISM
RESPIRATORY ACIDOSIS	Low pH (<7.35) + High PaCO ₂ (>45 mmHg).	Hypoventilation, COPD, opioid overdose, atelectasis, severe pneumonia.	Kidneys reabsorb bicarbonate (HCO ₃ ⁻).
RESPIRATORY ALKALOSIS	High pH (>7.45) + Low PaCO ₂ (<35 mmHg).	Hyperventilation, anxiety, panic attack, mechanical ventilator over-ventilation.	Kidneys excrete bicarbonate (HCO ₃ ⁻).
METABOLIC ACIDOSIS	Low pH (<7.35) + Low HCO ₃ (<22 mEq/L).	Diabetic Ketoacidosis (DKA), lactic acidosis, renal failure, severe diarrhea.	Lungs hyperventilate (Kussmaul respirations / blow off CO ₂).
METABOLIC ALKALOSIS	High pH (>7.45) + High HCO ₃ (>26 mEq/L).	Prolonged nasogastric suctioning, severe vomiting, diuretic overuse.	Lungs hypoventilate (retain CO ₂).

MECHANICAL VENTILATOR ALARM TROUBLESHOOTING (DOPE MNEMONIC)

HIGH-PRESSURE ALARM TRIGGERS

- Indicates obstruction to airflow or increased resistance.
- **Causes:** Patient biting ETT, secretions/mucus plug, kinked tubing, bronchospasm, pneumothorax, coughing.

LOW-PRESSURE ALARM TRIGGERS

- Indicates disconnect or leak in ventilator circuit.
- **Causes:** Tubing disconnection, extubation, cuff leak on endotracheal tube.



ABG & Ventilator Speed Check

What does an arterial blood gas showing pH 7.28, PaCO₂ 52 mmHg indicate?

Respiratory Acidosis (Low pH + High PaCO₂)

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GI ASSESSMENT & ENDOSCOPIC INVESTIGATIONS

⚡ **GI ASSESSMENT & ENDOSCOPY: Systematic abdominal exam: Inspection → Auscultation (before palpation!) → Percussion → Palpation. Endoscopic procedures require pre-op informed consent and post-op gag reflex verification!**

DIAGNOSTIC PROCEDURE	CLINICAL INDICATION & PROCEDURAL PROTOCOL	POST-PROCEDURE NURSING PRIORITY
UPPER GI ENDOSCOPY (EGD)	Visualizes esophagus, stomach, and duodenum. Diagnoses ulcers, gastritis, and varices.	Keep NPO pre-op. Verify gag reflex before offering food or water post-op to prevent aspiration.
COLONOSCOPY	Visualizes entire large intestine. Requires aggressive pre-op bowel preparation (polyethylene glycol / GoLYTELY).	Monitor for rectal bleeding, severe abdominal distension, and signs of bowel perforation (peritonitis).
ERCP	Endoscopic Retrograde Cholangiopancreatography. Visualizes biliary and pancreatic ducts using fluoroscopy.	Monitor closely for severe post-op complication: Acute Pancreatitis (severe epigastric pain radiating to back).

ABDOMINAL ASSESSMENT SEQUENCE

- • Always **AUSCULTATE BEFORE PALPATION OR PERCUSSION!**
- • Palpating or percussing first alters bowel motility and false-reports hyperactive or hypoactive bowel sounds.

COLONOSCOPY PERFORATION TRAP

Following diagnostic colonoscopy, sudden severe abdominal rigidity, rebound tenderness, and fever indicate **bowel perforation and peritonitis**. Report stat!



GI Assessment Speed Check

Why must you auscultate the abdomen before palpating during a physical exam?

Palpation alters bowel motility and distorts true bowel sound assessment

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UPPER & LOWER GI BLEEDING & IBD

 **GI BLEEDING:** **Upper GI Bleed** presents with Hematemesis (bright red or "coffee-ground") and Melena (black tarry stools); **Lower GI Bleed** presents with Hematochezia (bright red rectal bleeding)! Hypovolemic shock is primary risk.

GI BLEED SOURCE	COMMON ANATOMICAL CAUSES	STOOL / VOMIT PRESENTATION	EMERGENCY ACTION
UPPER GI BLEED (Above ligament of Treitz)	Peptic ulcer disease, esophageal varices, Mallory-Weiss tear, gastritis.	Hematemesis (bright red or coffee-ground emesis) and Melena (tarry black foul-smelling stools).	Large-bore IV access, type & crossmatch, blood transfusion, gastric lavage, endoscopy.
LOWER GI BLEED (Below ligament of Treitz)	Diverticular disease, hemorrhoids, inflammatory bowel disease, colorectal polyps/cancer.	Hematochezia (bright red blood mixed with or coating stool).	Hemodynamic stabilization, colonoscopy, angiography.

INFLAMMATORY BOWEL DISEASE (CROHN'S VS ULCERATIVE COLITIS)

CROHN'S DISEASE

- Transmural inflammation affecting **ANY part of GI tract** (mouth to anus).
- Characterized by **"Skip Lesions"** and cobblestone appearance.
- High risk of fistulas, strictures, and malabsorption.

ULCERATIVE COLITIS (UC)

- Mucosal inflammation confined strictly to the **Colon and Rectum**.
- Continuous ulceration starting at rectum.
- Severe bloody diarrhea and mucus; high risk of toxic megacolon.



GI Bleeding Speed Check

What stool characteristic is classically associated with an Upper GI Bleed?

Melena (Black, tarry, foul-smelling stools due to digested blood)

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PEPTIC ULCER DISEASE & GASTRIC DISORDERS

⚡ **PEPTIC ULCER DISEASE (PUD):** Erosion of mucosal lining caused by **Helicobacter pylori** infection or chronic NSAID use. **Gastric ulcers cause pain immediately after eating; Duodenal ulcers cause pain 2–3 hours post-meal!**

ULCER TYPE	PAIN TIMING & CHARACTERISTICS	PATIENT BODY HABITUS & RISK	PRIMARY ETIOLOGY
GASTRIC ULCER	Pain occurs immediately or 30–60 mins after meals . Food often *worsens* pain. Weight loss common due to food fear.	Normal or malnourished weight; high risk of gastric malignancy.	*H. pylori* (70%), NSAIDs, smoking.
DUODENAL ULCER	Pain occurs 2–3 hours after meals and during the night. Food or antacids *relieve* pain!	Well-nourished; pain awakens patient at night. Most common ulcer type.	*H. pylori* (90%), gastric acid hypersecretion.

PERFORATION EMERGENCY

- Peptic ulcer eroding through gastric/duodenal wall spills gastric contents into sterile peritoneum.
- **Classic Sign:** Sudden, severe, agonizing **board-like rigid abdomen**, rebound tenderness, and silent abdomen.
- ✓ *Action:* NPO, NG suction, emergency surgery!

MEDICAL MANAGEMENT

- Antibiotic triple/quadruple therapy to eradicate *H. pylori*.
- Proton Pump Inhibitors (PPIs: Pantoprazole, Omeprazole).
- H2 receptor blockers (Famotidine) and antacids.
- Avoid NSAIDs and alcohol.



PUD Speed Check

What physical sign indicates a perforated peptic ulcer peritonitis emergency?

Board-like rigid abdomen with severe knife-like pain and rebound tenderness

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APPENDICITIS, INTESTINAL OBSTRUCTION & PERITONITIS

🚑 **APPENDICITIS & OBSTRUCTION:** Acute appendicitis pain localizes at **McBurney's Point (RLQ)**. Rebound tenderness = Peritoneal irritation. Intestinal obstruction causes colicky pain, vomiting, and absolute constipation!

CONDITION	PATHOPHYSIOLOGICAL HALLMARK & CLINICAL SIGNS	CRITICAL NURSING & MEDICAL INTERVENTION
ACUTE APPENDICITIS	Obstruction of appendiceal lumen. Pain starts periumbilical, migrates to RLQ at McBurney's Point . Positive Rovsing's, Psoas, and Obturator signs.	NPO, IV fluids, prepare for appendectomy. Never give enemas or heating pads (causes rupture and peritonitis!).
INTESTINAL OBSTRUCTION	Mechanical (adhesions, hernia, tumors) or Functional (paralytic ileus) blockage. Colicky cramping pain, projectile vomiting, and absolute constipation.	NPO, Insert NG tube for gastric decompression, IV fluid resuscitation, monitor for bowel strangulation.
PERITONITIS	Inflammation of peritoneum from perforated appendix, ulcer, or bowel. Rigid board-like abdomen, fever, tachycardia, shock.	Emergency exploratory laparotomy, broad-spectrum IV antibiotics, fluid resuscitation.

APPENDICITIS HEAT PACK TRAP

Applying a heating pad to the abdomen of a patient with suspected appendicitis increases blood flow and tissue metabolism, causing **rapid appendiceal rupture and fatal peritonitis**. Strictly contraindicated!

⚡ MCBURNEY'S POINT:

Located one-third the distance from the Anterior Superior Iliac Spine (ASIS) to the umbilicus. Maximum tenderness in acute appendicitis!

🎯 **Obstruction & Appendicitis Speed Check**
Where is McBurney's point anatomically located on the abdomen? One-third distance from right ASIS to umbilicus (RLQ)

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NUR-08 Batch 2: Units IV, V & VI
(Cardiopulmonary & GI)

LIVER CIRRHOSIS, PORTAL HTN & ENCEPHALOPATHY

LIVER CIRRHOSIS: Irreversible hepatic scarring replacing normal liver architecture. **Portal Hypertension leads to Ascites, Esophageal Varices & Hepatic Encephalopathy (elevated ammonia levels)!**

CIRRHOTIC COMPLICATION	PATHOPHYSIOLOGICAL MECHANISM & CLINICAL SIGNS	CRITICAL NURSING INTERVENTIONS
PORTAL HYPERTENSION & ASCITES	High pressure in portal venous system + hypoalbuminemia (low oncotic pressure) leaks fluid into peritoneal cavity.	Abdominal distension, shifting dullness, fluid wave, risk of Spontaneous Bacterial Peritonitis (SBP). Low-sodium diet, diuretic therapy (Spironolactone/Furosemide), periodic Paracentesis .
ESOPHAGEAL VARICES	Collateral blood vessels form in lower esophagus due to portal backpressure. Thin-walled vessels prone to rupture.	Massive hematemesis, melena, hemorrhagic shock. Administer Octreotide / Vasopressin, prepare for endoscopic banding or Sengstaken-Blakemore tube .
HEPATIC ENCEPHALOPATHY	Inability of liver to detoxify nitrogenous waste leads to elevated blood ammonia levels crossing blood-brain barrier.	Asterixis ("liver flap"), fetor hepaticus (musty breath), confusion, lethargy, coma. Administer Lactulose (excretes ammonia via stools) and low-protein diet if indicated.

PARACENTESIS POSITIONING & BLADDER TRAP

- Patient must void **immediately before paracentesis** to prevent accidental trocar puncture of a distended urinary bladder.
- Position client upright in high-Fowler's position at bedside during tap.

⚡ ASTERIXIS (LIVER FLAP):

Inability to maintain dorsiflexion of hands when wrists are extended; hands flap involuntarily. Pathognomonic sign of hepatic encephalopathy!



Liver Cirrhosis Speed Check

What medication is administered orally to reduce serum ammonia in hepatic encephalopathy?

Lactulose (Creates osmotic diarrhea, trapping and excreting ammonia)

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PANCREATITIS, CHOLECYSTITIS & GASTRIC TUBES

⚡ **PANCREATITIS & BILIARY TRACT:** Acute pancreatitis involves auto-digestion of pancreas by premature activation of pancreatic enzymes. **Classic sign:** Severe epigastric pain radiating straight to the back, relieved by leaning forward!

DIGESTIVE DISORDER	ETIOLOGY & PATHOPHYSIOLOGICAL FEATURES	KEY CLINICAL SIGNS & NURSING CARE
ACUTE PANCREATITIS	Gallstones or chronic alcoholism trigger premature activation of pancreatic enzymes (trypsin), digesting the gland.	Severe boring epigastric pain radiating to back, elevated serum amylase & lipase. Grey Turner's & Cullen's signs (hemorrhage). NPO + NG tube.
CHOLECYSTITIS & GALLSTONES	Obstruction of cystic or common bile duct by gallstones (cholelithiasis), leading to inflammation.	Right upper quadrant (RUQ) pain radiating to right scapula/shoulder, nausea, jaundice. Positive Murphy's Sign (inspiratory arrest on palpation).

GASTRIC DECOMPRESSION & ENTERAL TUBE FEEDING SAFETY

NG TUBE VERIFICATION RULES

- **Gold Standard:** Abdominal X-ray confirmation of tip position prior to initiating feeds.
- Aspirate gastric contents (pH test < 5.5).
- *Never rely on auscultation (air injection) alone!*

TUBE FEEDING SAFETY

- Elevate head of bed to **at least 30°–45°** during and 1 hour post-feed to prevent aspiration.
- Check gastric residual volumes (GRV) q4h.
- Flush tube with 30 mL water before/after meds.



Pancreatitis & GI Tubes Speed Check

What physical sign indicates acute cholecystitis during abdominal palpation?

Murphy's Sign (Inspiratory arrest upon deep RUQ palpation)

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NUR-08 Batch 2: Units IV, V & VI
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CARDIOVASCULAR ASSESSMENT & ECG INTERPRETATION

 **CARDIOVASCULAR ASSESSMENT:** Systematic evaluation of cardiac output and tissue perfusion. **ECG Waveform:** P wave (atrial depolarization), QRS complex (ventricular depolarization), T wave (ventricular repolarization)!

CARDIOVASCULAR PARAMETER	CLINICAL ASSESSMENT TECHNIQUE & NORMAL FINDING	PATHOLOGICAL FINDING & SIGNIFICANCE
PULSES & PERFUSION	Palpate peripheral pulses (carotid, radial, femoral, dorsalis pedis) graded 0 to 4+. Assess capillary refill (< 2 seconds).	Weak thready pulses indicate low cardiac output or hypovolemic shock. Absent pulses indicate arterial occlusion.
JUGULAR VENOUS PRESSURE (JVP)	Assess jugular vein distension with patient elevated at 30°–45° angle.	Elevated JVP (> 3–4 cm above sternal angle) indicates Right-Sided Heart Failure or Cardiac Tamponade .
HEART SOUNDS	Auscultate S1 ("lub" / AV valves close) and S2 ("dub" / Semilunar valves close) across 5 auscultatory landmarks (APETM).	• S3 Gallop: Early sign of heart failure / fluid overload. • S4 Gallop: Stiff ventricular wall (hypertension/ LVH). • Murmurs indicate turbulent valve flow.

CARDIAC DIAGNOSTIC INVESTIGATIONS

ECHOCARDIOGRAM & STRESS TEST

- **Echocardiogram:** Ultrasound assessing heart valve structure and Ejection Fraction (EF normal: 55%–70%).
- **Cardiac Stress Test:** Evaluates coronary perfusion during physical exertion (treadmill) or pharmacologically (Dobutamine).

CARDIAC CATHETERIZATION (ANGIOGRAPHY)

- Gold standard for visualizing coronary artery stenosis.
- ***Post-op care:*** Monitor insertion site (femoral/radial) for bleeding, check distal pulses q15m, keep limb straight!



CV Assessment Speed Check

What is the normal adult ejection fraction (EF) range on an echocardiogram?

55% to 70%

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(Cardiopulmonary & GI)

MANDATORY BCLS / BLS MODULE (AHA STANDARDS)

 **MANDATORY BCLS / BLS MODULE: Basic Cardiopulmonary Life Support protocol for adult cardiac arrest. High-Quality CPR: Rate 100–120/min, Depth ≥ 2 inches (5 cm), Ratio 30:2, Minimal interruptions (< 10s)!**

BCLS STEP / METRIC	AHA STANDARDIZED PROCEDURAL PROTOCOL	CRITICAL RATIONALE & SAFETY RULE
1. SCENE SAFETY & CHECK	Ensure scene is safe. Tap shoulder and shout: "Are you okay?" Check breathing for max 10 seconds.	Never initiate CPR on an unsafe scene or responsive victim.
2. ACTIVATE EMS & AED	Call 112 / activate emergency response system and fetch AED immediately.	Rapid defibrillation is vital for shockable rhythms (VF/VT).
3. CHEST COMPRESSIONS	Heel of hand on lower half of sternum. Rate: 100–120/min . Depth: ≥ 2 inches (5 cm) . Allow full recoil.	Minimize interruptions! Never pause compressions > 10 seconds.
4. AIRWAY & BREATHING	Open airway via Head-Tilt/Chin-Lift (or Jaw-Thrust if trauma suspected). Deliver 2 rescue breaths (1 sec each) via bag-valve-mask.	Ratio is strictly 30 Compressions to 2 Breaths .

AED OPERATION & FOREIGN BODY AIRWAY OBSTRUCTION (CHOKING)

AED OPERATION PROTOCOL

- 1. Power on AED immediately.
- 2. Attach pads to bare chest (Upper right, lower left).
- 3. Clear victim during rhythm analysis.
- 4. If shock advised, deliver shock, then resume CPR instantly!

CHOKING RESCUE (HEIMLICH)

- • **Conscious Adult:** Abdominal thrusts inward and upward until object expels or victim collapses.
- • **Unconscious Adult:** Lower to floor, begin chest compressions (do not check pulse first). Look in mouth before breaths.



BCLS Speed Check

What is the chest compression-to-ventilation ratio for adult single-rescuer CPR?

30 Compressions to 2 Rescue Breaths (30:2)

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CORONARY ARTERY DISEASE & MYOCARDIAL INFARCTION

 **MYOCARDIAL INFARCTION (MI):** Irreversible myocardial necrosis caused by prolonged ischemia from coronary artery occlusion. **Emergency MONA Protocol + 12-lead ECG within 10 minutes!** STEMI requires primary PCI within 90 minutes.

CLINICAL CONDITION	PATHOPHYSIOLOGICAL HALLMARK	CLINICAL PRESENTATION & ECG	EMERGENCY PROTOCOL
STABLE ANGINA	Fixed atherosclerotic plaque narrows coronary artery; ischemia triggered by exertion, relieved by rest/nitrates.	Predictable chest discomfort lasting < 15 mins; ST depression on ECG.	Rest, Sublingual Nitroglycerin.
STEMI (Acute Myocardial Infarction)	Complete obstruction of coronary artery causing prolonged ischemia leading to irreversible myocardial necrosis.	Unrelenting chest discomfort lasting > 20 mins; ST-segment depression or ST-segment elevation on ECG.	MONA + Primary PCI (<90 mins) or Thrombolysis

ECG INFARCT LOCALIZATION & CARDIAC BIOMARKERS

INFARCT LOCALIZATION (ECG LEADS)

- **Anterior Wall MI:** Leads V1 to V4 (LAD artery / "Widow Maker").
- **Inferior Wall MI:** Leads II, III, aVF (Right Coronary Artery / RCA). *Caution: Watch for RV infarction!*
- **Lateral Wall MI:** Leads I, aVL, V5, V6 (Circumflex artery).

CARDIAC BIOMARKERS

- **Cardiac Troponin I & T:** Gold standard! Rises in 3–6 hrs, peaks at 24 hrs, remains elevated for **10–14 days**.
- **CK-MB:** Rises in 4–6 hrs, normalizes in 48–72 hrs.



MI Speed Check

What is the gold standard cardiac biomarker for diagnosing myocardial infarction?

Cardiac Troponin I and T (Elevated for 10–14 days)

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NUR-08 Batch 2: Units IV, V & VI
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HEMATOLOGIC ASSESSMENT, CBC & COAGULATION PROFILE

HEMATOLOGIC ASSESSMENT: Core blood parameters. **RBC indices (MCV, MCH, MCHC) differentiate microcytic vs macrocytic anemias; Coagulation profile (PT/INR for extrinsic/Warfarin; aPTT for intrinsic/Heparin)!**

LABORATORY TEST	STANDARD NORMAL RANGE	CRITICAL CLINICAL PANIC VALUES	PRIMARY NURSING ALERT
HEMOGLOBIN (Hb)	• Male: 14–18 g/dL • Female: 12–16 g/dL	< 7.0 g/dL (Transfusion threshold!) or > 20 g/dL.	Evaluate fatigue, tachycardia, orthostatic hypotension.
PLATELETS (PLT)	150,000 – 450,000 /mm ³	< 20,000 /mm³ (Risk of spontaneous fatal intracranial hemorrhage!).	Initiate Bleeding Precautions (soft toothbrush, no IM injections, fall precautions).
TOTAL WBC COUNT	4,500 – 11,000 /mm ³	< 2,000 /mm ³ (severe leukopenia) or > 30,000 /mm ³ (leukemoid / leukemia).	Monitor absolute neutrophil count (ANC). Screen for infection.
HEMATOCRIT (PCV)	• Male: 42%–52% • Female: 37%–47%	< 20% (severe hypoxemia) or > 60% (hyperviscosity / polycythemia).	Approximately 3 times the hemoglobin level (\$Hb \times 3\$).

COAGULATION CASCADE MONITORING & ANTICOAGULANT TARGETS

PT / INR (EXTRINSIC PATHWAY)

- Normal PT: **11–13.5 seconds**.
- Normal INR: **0.8–1.1**.
- **Therapeutic Warfarin (Coumadin) Target: INR 2.0 to 3.0** (2.5–3.5 for mechanical prosthetic heart valves).
- **Warfarin Antidote: Vitamin K (Phytonadione)!**

APTT (INTRINSIC PATHWAY)

- Normal aPTT: **30–40 seconds**.
- **Therapeutic Heparin Target: 1.5 to 2.5 times baseline control (60–80 seconds)!**
- If aPTT > 100 seconds: Turn off heparin infusion immediately and notify physician!
- **Heparin Antidote: Protamine Sulfate!**

SPONTANEOUS BLEEDING THRESHOLD

When platelet count falls below **20,000 /mm³**, the client is at imminent risk for spontaneous, life-threatening cerebral or gastrointestinal bleeding. Avoid all invasive procedures, suppositories, and razors!



Hematology Speed Check

What is the therapeutic INR target range for a client receiving oral Warfarin therapy? **INR 2.0 to 3.0 (2.5 to 3.5 for mechanical prosthetic heart valves)**

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BONE MARROW ASPIRATION & BIOPSY NURSING CARE

 **BONE MARROW ASPIRATION:** Definitive diagnostic test for leukemias, aplastic anemia, and multiple myeloma. **#1 Preferred anatomical site in adults: Posterior Superior Iliac Spine (PSIS)!** Sternum is used for aspiration only, never trephine biopsy.

BIOPSY PHASE	PROCEDURAL PROTOCOL & ANATOMICAL STANDARDS	CRITICAL NURSING RESPONSIBILITY
1. PRE-PROCEDURE	Verify written informed consent, check coagulation profile (PT/INR, platelets), provide emotional reassurance, position client.	Assess bleeding diathesis; administer mild pre-procedure sedative as prescribed.
2. SITE & POSITION	• Posterior Superior Iliac Spine (PSIS): Prone or lateral decubitus position. Safest adult site. • Sternum: Supine. (Risk of cardiac/aortic puncture!).	Ensure client remains completely motionless during needle penetration to prevent visceral injury.
3. INTRA-PROCEDURE	Surgeon infiltrates local anesthetic down to periosteum; Jamshidi needle inserted with rotational twisting pressure.	Warn client they will feel a sharp, brief pulling pain/ache during marrow aspiration!
4. POST-PROCEDURE	Apply firm manual pressure over biopsy puncture site for 5 to 10 minutes . Apply sterile pressure dressing.	Position client in supine position directly on puncture site for 30–60 minutes to provide natural compression hemostasis!

POST-BIOPSY BLEEDING ASSESSMENT

- Inspect pressure dressing for bleeding, hematoma formation, or excessive bruising every 15 mins for the first hour.
- Instruct client to avoid strenuous exercise, heavy lifting, or contact sports for **24 to 48 hours**.

STERNAL BIOPSY CONTRAINDICATION

A trephine core biopsy needle (cutting cylinder) is **STRICTLY CONTRAINDICATED IN THE STERNUM** due to the thin bone plate and catastrophic risk of puncturing the underlying ascending aorta or heart! Use PSIS instead.

**Bone Marrow Speed Check**

What is the preferred client position immediately following posterior iliac bone marrow biopsy?

Supine lying directly on the puncture site for 30–60 minutes to maintain hemostasis



ANEMIAS MASTER & POLYCYTHEMIA VERA

ANEMIA CLASSIFICATION: Reduced oxygen-carrying capacity of blood. **Microcytic Hypochromic (Iron Deficiency); Macrocytic Normochromic (Megaloblastic: B12 & Folate deficiency); Normocytic Normochromic (Aplastic Anemia / Bone marrow failure)!**

ANEMIA CLASS	ETIOLOGY & LABORATORY INDICES	PATHOGNOMONIC CLINICAL HALLMARKS	CORE NURSING INTERVENTIONS
IRON DEFICIENCY (IDA) (Most Common Worldwide)	Chronic blood loss, poor dietary intake, increased demand (pregnancy). • Microcytic, Hypochromic (Low MCV < 80 fL, Low Ferritin).	• Koilonychia (spoon-shaped concave brittle nails). • Pica (craving dirt/ice/clay). • Cheilosis & smooth sore tongue.	Oral Ferrous Sulfate with Vitamin C / Orange Juice (enhances absorption); use straw for liquid iron (prevents tooth staining!).
PERNICIOUS / VIT B12 DEFICIENCY	Autoimmune destruction of gastric parietal cells \$ ightarrow\$ Lack of Intrinsic Factor (IF) or ileal resection. • Macrocytic (High MCV > 100 fL).	• Beefy red smooth tongue (Glossitis) . • Neurological deficits: Paresthesias, loss of vibratory sense, ataxia! (Folate lacks neuro signs).	Lifelong monthly Intramuscular Vitamin B12 (Cyanocobalamin) injections. (Oral B12 will not absorb without IF!).
APLASTIC ANEMIA	Autoimmune or toxic destruction of bone marrow stem cells (radiation, benzene, chloramphenicol).	PANCYTOPENIA: Severe anemia + Thrombocytopenia (bleeding) + Leukopenia (frequent severe infections).	Protective isolation (Neutropenic precautions), avoid all IM injections, prepare for Bone Marrow Transplant.

POLYCYTHEMIA VERA (PV)

- Chronic myeloproliferative disorder causing massive overproduction of RBCs (Hematocrit > 55%–60%).
- **Complication:** Extreme blood hyperviscosity \$ ightarrow\$ Thrombosis, stroke, MI, severe pruritus after warm baths.
- **Treatment:** Therapeutic Phlebotomy (blood removal) & Hydration!

FOLIC ACID IN B12 DEFICIENCY TRAP

Administering Folic Acid alone to a client with Vitamin B12 deficiency corrects the anemia on CBC, but **fails to halt irreversible neurological spinal cord degeneration (Subacute Combined Degeneration)!** Always confirm B12 levels first.

**Anemia Speed Check**

What physical sign distinguishes Vitamin B12 deficiency anemia from Folic Acid deficiency anemia?

Neurological symptoms (Paresthesias, ataxia, and peripheral neuropathy occur only in B12 deficiency)

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THALASSEMIA & SICKLE CELL CRISIS

HEMOGLOBINOPATHIES: Autosomal recessive genetic disorders of hemoglobin synthesis. **Thalassemia Major (Cooley's Anemia)** requires chronic lifelong transfusions leading to **Iron Overload (Hemosiderosis)**; Sickle Cell Anemia triggers painful vaso-occlusive crisis!

HEMOGLOBINOPATHY	PATHOPHYSIOLOGICAL DEFECT	CLASSIC CLINICAL MANIFESTATIONS	CLINICAL NURSING PRIORITY
THALASSEMIA MAJOR (Cooley's Anemia / α -thal)	Defective synthesis of α -globin chains leads to unstable RBCs, severe chronic hemolysis, and ineffective erythropoiesis.	- Severe microcytic anemia (Hb 4–6 g/dL). Chipmunk Facies (maxillary hypertrophy due to bone marrow expansion). - Hepatosplenomegaly, bronze skin.	Lifelong regular blood transfusions every 2–4 weeks + Daily Iron Chelation Therapy (Deferasirox / Deferoxamine) to prevent iron overload!
SICKLE CELL DISEASE (SCD)	Point mutation replacing glutamic acid with valine on β -chain (HbS). Deoxygenation causes RBCs to sickle into rigid crescent shapes.	Vaso-Occlusive Crisis (VOC): Agonizing bone/joint pain, acute chest syndrome. - Splenic sequestration, avascular necrosis.	HYDRATION (IV fluids), OXYGENATION, and AGGRESSIVE PAIN RELIEF (IV Opioids)!

HEMOSIDEROSIS & IRON CHELATION IN THALASSEMIA

HEMOSIDEROSIS (IRON OVERLOAD)

- Each unit of PRBCs deposits ~200–250 mg of elemental iron in tissues.
- The body lacks a physiological mechanism to excrete excess iron.
- Iron accumulates in **Heart (Cardiomyopathy / #1 cause of death!)**, Liver (cirrhosis), and Pancreas (Bronze Diabetes).

IRON CHELATION AGENTS

- Deferoxamine (Desferal):** Subcutaneous or IV infusion over 8–12 hours daily. Causes reddish urine.
- Deferasirox (Exjade):** Oral chelator taken once daily.
- Target: Maintain serum ferritin < 1,000 ng/mL.

SICKLE CELL COLD COMPRESS TRAP

Applying ice packs or cold compresses to painful swollen joints during a Sickle Cell vaso-occlusive crisis causes **severe vasoconstriction, accelerating local sickling and tissue infarction!** Always apply warm, soothing compresses.



Hemoglobinopathies Speed Check

What is the primary cause of mortality in chronically transfused Thalassemia Major clients?

Cardiac Hemosiderosis (Iron overload inducing fatal heart failure and arrhythmias)



BLEEDING DISORDERS: HEMOPHILIA, ITP & DIC

COAGULATION DISORDERS: Hemophilia A (Factor VIII deficiency, X-linked recessive); Hemophilia B (Factor IX deficiency / Christmas disease). **Classic manifestation is Hemarthrosis (bleeding into joints).** DIC is a consumption coagulopathy causing simultaneous thrombosis and massive hemorrhage!

BLEEDING DISORDER	DEFECTIVE CLOTTING MECHANISM	DIAGNOSTIC LABORATORY CLUES	EMERGENCY TREATMENT
HEMOPHILIA A (Classic / 80%)	Deficiency of Clotting Factor VIII . X-linked recessive (affects males; carrier females).	Prolonged aPTT ; normal PT/INR, normal platelet count.	IV Factor VIII Concentrate infusion, DDAVP (Desmopressin) for mild cases.
HEMOPHILIA B (Christmas Disease)	Deficiency of Clotting Factor IX . X-linked recessive inheritance.	Prolonged aPTT ; normal PT and platelets.	IV Factor IX Concentrate infusion.
IMMUNE THROMBOCYTOPENIC PURPURA (ITP)	Autoimmune anti-platelet antibodies destroy platelets in the spleen → profound thrombocytopenia.	Platelets < 20,000–50,000 /mm ³ ; normal coagulation factors. Petechiae, purpura.	Corticosteroids (Prednisone), IVIG, Splenectomy for refractory cases.
DIC (Disseminated Intravascular Coagulation)	Widespread activation of coagulation cascade leading to microvascular thrombosis and consumption of clotting factors.	Prolonged PT/aPTT, Low Platelets, Low Fibrinogen, Sky-High D-Dimer	Treat underlying trigger (Sepsis/Abruptio)

HEMARTHROSIS EMERGENCY MANAGEMENT (RICE)

- Acute bleeding into knees, elbows, or ankles causes severe pain and joint deformity.
- **R = Rest:** Immobilize joint immediately.
- **I = Ice:** Apply cold packs (vasoconstriction).
- **C = Compression:** Elastic bandage.
- **E = Elevation:** Elevate above heart level. *Active ROM is strictly forbidden during acute bleeding!*

ASPIRIN CONTRAINDICATION TRAP

Administering Aspirin, Ibuprofen, or other NSAIDs to a client with Hemophilia or ITP for pain relief is **FATAL!** NSAIDs permanently inhibit platelet aggregation and trigger uncontrolled hemorrhagic bleeding. Use Acetaminophen (Paracetamol) or prescribed opioids instead.

**Coagulation Speed Check**

Which clotting factor is deficient in classical Hemophilia A?

Clotting Factor VIII (Anti-hemophilic factor)



LEUKEMIAS & NEUTROPENIC PRECAUTIONS

 **LEUKEMIAS:** Malignant proliferation of immature non-functional white blood cells (blasts) crowding out normal bone marrow elements. **Absolute Neutrophil Count (ANC) < 500 /mm³ indicates severe neutropenic infection emergency!**

LEUKEMIA TYPE	TARGET DEMOGRAPHIC	PATHOLOGICAL HALLMARK & GENETICS	CLINICAL FEATURES
ACUTE LYMPHOBLASTIC (ALL)	Children (Peak age 2–5 years); most common pediatric leukemia.	Proliferation of malignant lymphoblasts. High cure rate (> 85% in pediatric cohorts).	Bone pain, pallor, petechiae, fever, hepatosplenomegaly.
ACUTE MYELOID (AML)	Adults (incidence rises steeply with age > 60 years).	Proliferation of immature myeloblasts; pathognomonic Auer Rods inside blast cytoplasm!	Fatigue, severe opportunistic infections, severe bleeding.
CHRONIC MYELOID (CML)	Middle-aged adults (40–60 years).	Presence of the Philadelphia Chromosome, t(9;22) translocation forming BCR-ABL tyrosine kinase.	Massive splenomegaly, fatigue; treated with Imatinib (Gleevec) .
CHRONIC LYMPHOCYTIC (CLL)	Elderly adults (> 65 years); indolent slow course.	Proliferation of mature-appearing incompetent B-lymphocytes; Smudge Cells on blood smear.	Often asymptomatic; generalized painless lymphadenopathy.

NEUTROPENIC PRECAUTIONS & INFECTION SHIELD (ANC < 1,000 /MM³)

CALCULATING ANC (FORMULA)

- $\text{ANC} = \text{Total WBC} \times \frac{\text{Segmented Neutrophils} + \% \text{Bands}}{100}$
- Normal ANC: 1,500 to 8,000 /mm³.
- **Moderate Neutropenia:** 500 to 1,000 /mm³.
- **Severe Neutropenia:** < 500 /mm³ (Extreme risk of fatal bacteremia!).

PROTECTIVE ISOLATION PROTOCOL

- Private room with positive pressure HEPA filtration.
- Strict hand hygiene before entering.
- **Zero fresh flowers, potted plants, or standing water** (harbors "Pseudomonas aeruginosa").
- **Zero raw fruits, vegetables, or unpasteurized dairy** (low-bacterial cooked diet only!).



Leukemia Speed Check

What genetic chromosomal abnormality is diagnostic of Chronic Myeloid Leukemia (CML)?

The Philadelphia Chromosome, t(9;22) translocation forming BCR-ABL gene



LYMPHOMAS & MULTIPLE MYELOMA (CRAB CRITERIA)

LYMPHOMAS & MYELOMA: Malignancies of the lymphoid and plasma cell systems. **Hodgkin Lymphoma** features **Reed-Sternberg Cells**; **Multiple Myeloma** features malignant plasma cell proliferation diagnosed via **CRAB** criteria and Bence-Jones proteinuria!

LYMPHOPROLIFERATIVE DISORDER	PATHOLOGICAL HALLMARK & CELLULAR ORIGIN	CLASSIC CLINICAL MANIFESTATIONS	STAGING / DIAGNOSTIC TEST
HODGKIN LYMPHOMA (HL)	Malignancy of B-lymphocytes characterized pathognomically by giant multinucleated Reed-Sternberg Cells ("Owl's eye" appearance).	• Painless, rubbery, enlarged cervical/supraclavicular lymph node. • B Symptoms: Unexplained fever, drenching night sweats, weight loss > 10%. • Alcohol-induced node pain!	Excisional lymph node biopsy; Ann Arbor staging system (Stages I to IV).
NON-HODGKIN LYMPHOMA (NHL)	Heterogeneous group of lymphoid malignancies lacking Reed-Sternberg cells. Arises from multiple peripheral lymph nodes with unpredictable spread.	Painless generalized lymphadenopathy, extranodal involvement (GI, bone, CNS), hepatosplenomegaly.	Biopsy, immunophenotyping (B-cell vs T-cell lymphoma).
MULTIPLE MYELOMA	Neoplastic proliferation of malignant plasma cells in bone marrow producing monoclonal paraprotein (M-Spike / Bence-Jones Protein).	Severe bone pain (punched-out lytic lesions in skull/spine), pathologic fractures, hypercalcemia, renal failure.	Serum protein electrophoresis (SPEP), 24-hr urine for Bence-Jones protein, marrow biopsy.

MULTIPLE MYELOMA "CRAB" DIAGNOSTIC CRITERIA

CRAB CRITERIA BREAKDOWN

- **C = Calcium Elevation:** Hypercalcemia (> 11 mg/dL) from bone breakdown.
- **R = Renal Impairment:** Serum creatinine > 2 mg/dL from light-chain cast nephropathy.
- **A = Anemia:** Normocytic anemia from bone marrow crowding.
- **B = Bone Lesions:** Punched-out osteolytic bone lesions on skeletal survey X-rays!

NURSING PRIORITIES IN MYELOMA

- **Aggressive Hydration:** 3 to 4 L/day of fluid to flush calcium and protein casts through kidneys!
- Safe patient transfers and fall prevention to avoid pathological fractures.
- Pain management and monitoring for spinal cord compression.



Lymphoma & Myeloma Speed Check

What pathognomonic cell type is identified on lymph node biopsy in Hodgkin Lymphoma?

Reed-Sternberg Cells (Giant binucleated cells resembling "owl's eyes")

BLOOD TRANSFUSION PROTOCOL & COMPONENT THERAPY

BLOOD TRANSFUSION PROTOCOL: High-risk clinical procedure requiring strict verification. **Packed Red Blood Cells (PRBCs) must be infused within 4 HOURS** of leaving blood bank; **Platelets must never be refrigerated!** 2-Nurse bedside verification is mandatory.

BLOOD COMPONENT	VOLUME & COMPOSITION	STANDARD INFUSION DURATION	CLINICAL INDICATION
PACKED RED CELLS (PRBCs)	~250–300 mL (hematocrit ~65%–70%; minimal plasma).	Infuse over 2 to 4 hours MAX! (Never exceed 4 hours due to bacterial proliferation!).	Severe anemia (Hb < 7–8 g/dL), acute hemorrhage, symptomatic hypoxemia.
PLATELETS	Single donor or pooled unit (200–300 mL). Stored at room temperature (20°C–24°C) with constant agitation!	Infuse rapidly over 15 to 30 minutes per unit using a specialized platelet filter.	Thrombocytopenia (platelets < 10,000–20,000) or massive active bleeding.
FRESH FROZEN PLASMA (FFP)	~200–250 mL containing all coagulation factors (thawed in water bath).	Infuse within 24 hours of thawing over 30 to 60 minutes .	Coagulation factor deficiencies, massive transfusion protocol, Warfarin reversal.

THE 2-NURSE BEDSIDE VERIFICATION CHECKLIST

VERIFICATION MANDATES

- Two registered nurses must verify **at the patient's bedside**:
- 1. Patient full name and unique hospital ID number against wristband.
- 2. Blood unit bag serial number, blood type (ABO & Rh status), and crossmatch compatibility slip.
- 3. Expiration date and time of the blood unit.

INFUSION SETUP RULES

- Use dedicated IV cannula (minimum **18 to 20 gauge** to prevent RBC hemolysis).
- **0.9% Normal Saline is the ONLY compatible IV solution!** (Dextrose causes hemolysis; Ringer's causes clotting from calcium!).
- Prime blood tubing with blood filter.

4-HOUR BLOOD INFUSION EXPIRATION TRAP

A blood unit initiated at 12:00 PM is still infusing at 4:30 PM:

✗ FATAL INFECTION RISK! Blood hanging at room temperature beyond **4 hours** undergoes rapid bacterial growth. The nurse must stop the infusion immediately, discard the remaining blood, and replace tubing!

 Transfusion Protocol Speed Check

What is the only compatible intravenous crystalloid solution that can be co-infused with blood? **0.9% Normal Saline (0.9% NaCl)**



TRANSFUSION REACTIONS & EMERGENCY PROTOCOL

TRANSFUSION REACTIONS: Most reactions occur within the **FIRST 15 MINUTES (or first 50 mL)**. Stay with the patient during this period! Acute Hemolytic reaction caused by ABO incompatibility is an emergency. First action: **STOP THE INFUSION!**

REACTION TYPE	PATHOPHYSIOLOGICAL ETIOLOGY	CLINICAL MANIFESTATIONS	EMERGENCY ACTION
ACUTE HEMOLYTIC (Medical Catastrophe)	ABO or Rh incompatibility (clerical mislabeling error!). Antibodies attack donor RBCs & intravascular hemolysis.	• Flank / Low back pain! • Chills, fever, dyspnea. • Hypotension, red/brown urine (hemoglobinuria), shock!	STOP TRANSFUSION IMMEDIATELY! Infuse normal saline via new IV tubing, treat shock, notify blood bank.
FEBRILE NON-HEMOLYTIC (Most Common)	Antibodies directed against donor white blood cells or cytokines accumulating in stored blood.	Fever (elevation $\geq 1^{\circ}\text{C}$ / 2°F from baseline), chills, headache, flushing, anxiety.	Stop transfusion, notify physician, administer Antipyretics (Acetaminophen). Use leukocyte-poor filter next time.
ALLERGIC / ANAPHYLACTIC	Sensitivity to donor plasma proteins (IgA deficiency antibodies in recipient).	Urticaria, hives, flushing, pruritus. Severe: Stridor, bronchospasm, shock.	Stop blood. Give Antihistamines (Diphenhydramine); Epinephrine & steroids if anaphylaxis occurs.
CIRCULATORY OVERLOAD (TACO)	Infusing blood too rapidly into volume-sensitive patients (heart failure, renal failure).	Dyspnea, orthopnea, JVD, crackles in lung bases, hypertension.	Slow or stop infusion, sit patient upright with legs dependent, administer IV Furosemide.

THE 5-STEP EMERGENCY TRANSFUSION REACTION PROTOCOL

STEPS 1 TO 3: IMMEDIATE LIFE STABILIZATION

1. **STOP THE TRANSFUSION IMMEDIATELY!** (Do not simply slow it down!).
2. Disconnect blood tubing at IV hub; maintain IV access by hanging a **NEW bag of 0.9% Normal Saline with entirely NEW tubing!** (Never flush blood in line!).
3. Notify treating physician and blood bank stat; assess vitals q5m.

STEPS 4 TO 5: INVESTIGATION & DOCUMENTATION

4. Collect blood samples (hemolysis screen) and patient's **first voided urine specimen** (check for free hemoglobinuria!).
5. Return the remaining blood bag, attached tubing, and label paperwork to the blood bank for clerical investigation!

FLUSHING BLOOD TUBING TRAP

When a patient exhibits signs of a transfusion reaction, **NEVER flush the existing IV tubing with saline!** Flushing pushes the remaining 30–50 mL of incompatible blood trapped in the tubing straight into the patient's circulation, worsening fatal hemolysis!



Transfusion Reaction Speed Check

What is the absolute first nursing intervention when a transfusion reaction is suspected?

Stop the blood transfusion immediately and disconnect tubing at the IV hub



ENDOCRINE ASSESSMENT: DIABETES INSIPIDUS VS SIADH

 **DI VS SIADH: Disorders of Antidiuretic Hormone (ADH / Vasopressin) secretion by posterior pituitary. Diabetes Insipidus (DI) = Lack of ADH → Massive dilute polyuria & Hypernatremia; SIADH = Excess ADH → Water retention, Concentrated urine & Dilutional Hyponatremia!**

CLINICAL PARAMETER	DIABETES INSIPIDUS (DI) — "DRY INSIDE"	SIADH — "SOAKED INSIDE"
ADH Hormone Level	DEFICIENT / ABSENT ADH (Central trauma/surgery or Nephrogenic resistance).	EXCESSIVE / UNREGULATED ADH (Small cell lung cancer, head injury, stroke).
Urine Output & Color	Massive dilute polyuria (4 to 20 Liters/day!) . Pale, clear urine like water.	Severe oliguria with dark, concentrated, amber urine.
Urine Specific Gravity	DILUTE: < 1.005 (Urine osmolality < 200 mOsm/kg).	CONCENTRATED: > 1.030 (Urine osmolality > 800 mOsm/kg).
Serum Sodium (Na ⁺)	HYPERNATREMIA (> 145 mEq/L) (Hemoconcentration due to pure water loss).	DILUTIONAL HYPONATREMIA (< 135 mEq/L) (Cerebral edema & seizure risk!).
Primary Treatment	- Administer synthetic ADH: Desmopressin (DDAVP). - Generous fluid replacement (D5W / hypotonic saline).	STRICT FLUID RESTRICTION (500–1,000 mL/day). - Hypertonic 3% Saline IV for severe hyponatremia (<120).

SIADH SEIZURE PRECAUTIONS

- Rapid dilutional hyponatremia forces water into brain cells, causing cerebral edema and brainstem herniation!
- Implement seizure precautions: padded bed rails, suction apparatus, and emergency hypertonic 3% NaCl.

RAPID SODIUM CORRECTION TRAP

Correcting chronic hyponatremia in SIADH too rapidly (> 8–10 mEq/L in 24 hrs) causes catastrophic **Central Pontine Myelinolysis (Osmotic Demyelination Syndrome)**, resulting in irreversible spastic quadriplegia! Infuse 3% saline slowly via infusion pump.



DI vs SIADH Speed Check

What is the primary pharmacological management for central Diabetes Insipidus?

Desmopressin (DDAVP / synthetic Antidiuretic Hormone)



DIABETES MELLITUS: PATHOPHYSIOLOGY & DIAGNOSIS

 **DIABETES MELLITUS (DM):** Metabolic disorder of carbohydrate, protein, and fat metabolism. **Type 1** (Autoimmune destruction of pancreatic β -cells → Absolute insulin deficiency); **Type 2** (Insulin resistance + relative secretory defect). **Diagnostic:** HbA1c $\geq 6.5\%$!

DIAGNOSTIC CRITERION	ADA STANDARDIZED NUMERICAL THRESHOLD	CLINICAL TESTING STANDARD
GLYCATED HEMOGLOBIN (HbA1c)	$\geq 6.5\%$ (Normal < 5.7%; Pre-diabetes 5.7%–6.4%).	Reflects average blood glucose over preceding 2 to 3 months (90–120 days) ; fasting not required.
FASTING PLASMA GLUCOSE (FPG)	≥ 126 mg/dL (7.0 mmol/L) (Normal < 100 mg/dL).	Fasting defined as zero caloric intake for at least 8 hours prior to blood draw.
2-HR ORAL GLUCOSE (OGTT)	≥ 200 mg/dL (11.1 mmol/L).	Measured 2 hours following a 75 g anhydrous oral glucose load.
RANDOM PLASMA GLUCOSE	≥ 200 mg/dL accompanied by classic symptoms.	Classic Triad: Polyuria, Polydipsia, and Polyphagia + unexplained weight loss.

MAJOR ORAL ANTIDIABETIC DRUG CLASSES (OADS)

BIGUANIDES: METFORMIN (GLUCOPHAGE)

- First-line therapy for Type 2 DM. Decreases hepatic glucose production.
- Does not cause hypoglycemia; aids weight loss.
- **ALERT:** Hold Metformin **48 hours before and after IV contrast dye** to prevent fatal Lactic Acidosis!

SULFONYLUREAS: GLIPIZIDE / GLIMEPIRIDE

- Stimulates pancreatic β -cells to secrete insulin.
- **Major Side Effect:** High risk of prolonged **Hypoglycemia** and weight gain!
- Administer 30 minutes before meals. Avoid alcohol.

METFORMIN RADIOLOGICAL CONTRAST TRAP

A diabetic client taking Metformin is scheduled for a CT scan with IV iodinated contrast. Metformin must be **held on the day of procedure and for 48 hours post-scan** until serum creatinine confirms normal renal function!



Diabetes Speed Check

What diagnostic HbA1c threshold confirms a diagnosis of Diabetes Mellitus?

HbA1c $\geq 6.5\%$

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INSULIN THERAPY MASTER & SOMOGYI VS DAWN

 **INSULIN THERAPY MASTER:** Injected subcutaneously at 45°–90° angle. **Regular Insulin is the ONLY insulin that can be administered intravenously (IV)!** Rotate injection sites within one anatomical zone to prevent Lipodystrophy.

INSULIN CLASS	GENERIC NAMES	ONSET OF ACTION	PEAK ACTION PERIOD	DURATION OF ACTION
RAPID-ACTING	Lispro (Humalog), Aspart (Novolog), Glulisine	10 to 15 minutes	30 to 90 minutes (Highest hypoglycemia risk!)	3 to 5 hours
SHORT-ACTING	Regular Insulin (Humulin R, Novolin R)	30 to 60 minutes	2 to 4 hours	5 to 8 hours
INTERMEDIATE	NPH (Neutral Protamine Hagedorn)	1 to 2 hours	4 to 12 hours	14 to 24 hours
LONG-ACTING	Glargine (Lantus), Detemir (Levemir), Degludec	1 to 2 hours	NO PEAK! (Peakless flat plateau)	24 hours+

MORNING HYPERGLYCEMIA: SOMOGYI EFFECT VS DAWN PHENOMENON

SOMOGYI EFFECT (REBOUND HYPERGLYCEMIA)

- Excessive evening insulin causes **undetected hypoglycemia at 2:00–3:00 AM**.
- Counter-regulatory hormones (cortisol, glucagon) surge, causing rebound morning hyperglycemia!
- ✓ **Action:** Decrease bedtime insulin dose or add a bedtime carbohydrate snack!

DAWN PHENOMENON

- Normal physiological surge of growth hormone and cortisol around 4:00–8:00 AM causes morning hyperglycemia.
- **Blood glucose at 2:00–3:00 AM is NORMAL or HIGH** (not hypoglycemic!).
- ✓ **Action:** Increase evening/bedtime insulin dose!

CLEAR BEFORE CLOUDY MIXING TRAP

When mixing Regular (Clear) and NPH (Cloudy) insulin in one syringe:

✓ **MANDATORY SEQUENCE:** Inject air into NPH (cloudy), inject air into Regular (clear), **draw up REGULAR (clear) first**, then draw up NPH (cloudy)! (Mnemonic: *RN = Regular before NPH*).



Insulin Therapy Speed Check

Which is the only insulin type that can be safely administered intravenously (IV)? **Regular Insulin (Short-acting pure crystalline insulin)**

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NUR-08 Batch 3: Units VII, VIII & IX (Blood, Endocrine, Skin)

ACUTE DIABETIC EMERGENCIES: DKA VS HHS

 **DKA VS HHS: Life-threatening acute metabolic crises. DKA (Type 1 DM: Absolute insulin deficiency → Ketoacidosis, Kussmaul respirations, fruity breath); HHS (Type 2 DM: Severe hyperglycemia > 600 mg/dL, extreme hyperosmolality > 320 mOsm/kg, zero ketoacidosis)!**

CLINICAL PARAMETER	DIABETIC KETOACIDOSIS (DKA)	HYPERGLYCEMIC HYPEROSMOLAR STATE (HHS)
Typical Patient Population	Type 1 Diabetes Mellitus (Younger patients; sudden onset < 24 hrs).	Type 2 Diabetes Mellitus (Elderly patients; insidious onset over days/weeks).
Blood Glucose Level	Moderately elevated: 250 to 500 mg/dL.	EXTREMELY HIGH: > 600 to 1,200 mg/dL!
Arterial pH & Bicarbonate	METABOLIC ACIDOSIS: pH < 7.30, Serum HCO_3^- < 15 mEq/L.	NORMAL: pH > 7.30, Serum HCO_3^- > 18 mEq/L (No acidosis).
Ketones (Urine & Serum)	POSITIVE (Heavy ketonemia and ketonuria; fruity acetone breath).	NEGATIVE or TRACE (Minimal ketone production).
Serum Osmolality & Deficit	Variable (< 320 mOsm/kg); fluid deficit ~4 to 6 Liters.	HYPEROSMOLAR: > 320 mOsm/kg ; profound fluid deficit (8 to 12 Liters!).

EMERGENCY MANAGEMENT & POTASSIUM SHIFT DYNAMICS

1. REHYDRATION PROTOCOL

- First line is rapid infusion of **0.9% Normal Saline (1 L/hr initially)** to restore perfusion.
- When blood glucose drops to **250 mg/dL**, add **5% Dextrose (D5W)** to IV fluids to prevent cerebral edema and rapid hypoglycemia!

2. POTASSIUM (K^+) SAFETY RULE

- Insulin shifts potassium *into cells*, precipitating fatal **Hypokalemia!**
- **MANDATORY:** Verify serum K^+ ≥ 3.5 mEq/L before starting IV insulin infusion! If $\text{K}^+ < 3.5$, give potassium replacement first.

IV INSULIN POTASSIUM ARREST TRAP

Initiating an IV Regular Insulin drip on a DKA client with a baseline serum potassium of **3.1 mEq/L** causes rapid intracellular potassium shifting, resulting in sudden **fatal cardiac arrest!** Always replete potassium above 3.5 mEq/L first.

 **DKA vs HHS Speed Check**
What respiratory pattern is pathognomonic of severe Diabetic Ketoacidosis?

Kussmaul Respirations (Deep, rapid respirations attempting to blow off acidic CO_2)

CHRONIC COMPLICATIONS & DIABETIC FOOT CARE

⚡ **DIABETIC FOOT CARE:** Diabetic peripheral neuropathy + peripheral vascular disease leads to painless foot ulcers and gangrene. **Inspect feet daily with a mirror; never walk barefoot; cut toenails straight across!** Semmes-Weinstein monofilament test evaluates protective sensation.

COMPLICATION CLASS	PATHOPHYSIOLOGICAL TISSUE TARGET	CLINICAL MANIFESTATION & SCREENING
MICROVASCULAR	Thickening of capillary basement membranes in end-organs: Retinopathy, Nephropathy, Neuropathy.	• Nephropathy: Microalbuminuria (annual urine screen; ACE inhibitors protect kidneys!). • Retinopathy: Annual dilated eye exam.
MACROVASCULAR	Accelerated atherosclerosis of medium and large arteries.	Coronary Artery Disease (silent MI!), Peripheral Artery Disease (PAD / intermittent claudication), and Stroke.
DIABETIC NEUROPATHY	Demyelination of peripheral sensory nerves (distal symmetrical "stocking-glove" sensory loss).	Loss of protective sensation (painless blisters/cuts become infected osteomyelitis ulcers!).

PATIENT EDUCATION: RULES OF PREVENTIVE FOOT CARE

DAILY HYGIENE & INSPECTION

- Inspect entire feet and between toes daily using an unbreakable mirror.
- Wash daily with mild lukewarm soap/water; test water temp with elbow/thermometer (never use feet!).
- Pat dry thoroughly, especially **between toes**.
- Apply moisturizing lotion to tops and bottoms of feet, but **NEVER BETWEEN TOES** (prevents fungal maceration!).

FOOTWEAR & NAIL CARE RULES

- **NEVER walk barefoot**, even indoors!
- Wear clean, breathable, white cotton socks (shows blood/drainage).
- Inspect inside of shoes for pebbles/nails before putting them on.
- Cut toenails **STRAIGHT ACROSS**, never round corners (prevents ingrown toenails).
- Never use heating pads or hot water bottles on feet!



Foot Care Speed Check

Why must diabetic patients avoid applying moisturizing lotion between their toes? **Moisture between toes promotes skin maceration and fungal infections**



THYROID DISORDERS & THYROID STORM CRISIS

THYROID DISORDERS: Hyperthyroidism (Graves' disease: High T3/T4, Low TSH → Exophthalmos, tachycardia, heat intolerance); Hypothyroidism (Hashimoto's: Low T3/T4, High TSH → Weight gain, cold intolerance, myxedema)! Thyroid Storm is a hypermetabolic life-threatening crisis.

CLINICAL DOMAIN	HYPERTHYROIDISM (GRAVES' DISEASE)	HYPOTHYROIDISM (MYXEDEMA / HASHIMOTO'S)
Metabolic Rate & Weight	HYPERMETABOLIC: Weight loss despite ravenous appetite (polyphagia), Heat Intolerance , warm moist skin.	HYPOMETABOLIC: Weight gain despite poor appetite, Cold Intolerance , dry coarse scaly skin, brittle hair.
Cardiovascular & Vitals	Tachycardia, palpitations, systolic hypertension, atrial fibrillation.	Bradycardia, hypotension, hypothermia, pericardial effusion.
Neuromuscular & Mood	Nervousness, fine hand tremors, insomnia, emotional lability, Exophthalmos (bulging eyes) , goiter.	Lethargy, fatigue, depression, slurred speech, forgetfulness, periorbital edema, non-pitting facial edema.
Laboratory Profile	High Free T3/T4 + Suppressed TSH (< 0.1 mU/L).	Low Free T3/T4 + Elevated TSH (> 4.5 mU/L) (Primary thyroid failure).
Pharmacological Therapy	Antithyroid drugs: Methimazole , Propylthiouracil (PTU) , Beta-blockers (Propranolol), Radioactive Iodine (RAI).	Lifelong thyroid hormone replacement: Levothyroxine (Synthroid) taken in morning on empty stomach!

THYROID STORM (THYROTOXIC CRISIS) EMERGENCY PROTOCOL

THYROID STORM MANIFESTATIONS

- Triggered by infection, trauma, surgery, or stopping antithyroid meds.
- **Extreme hyperthermia (> 40°C / 104°F)**, severe tachycardia (> 140 bpm), delirium, heart failure, and shock!

EMERGENCY INTERVENTIONS

- 1. Administer **PTU / Methimazole** (blocks synthesis).
- 2. Give **iodine solution (Lugol's / SSKI)** 1 hour after PTU to block hormone release.
- 3. IV **Propranolol** (blocks beta-adrenergic symptoms).
- 4. IV **Hydrocortisone** + Cooling blankets (no aspirin!).

THYROID STORM ASPIRIN CONTRAINDICATION

✗ **FATAL MISTAKE:** Giving Aspirin (acetylsalicylic acid) to lower fever during Thyroid Storm.

✓ **RATIONALE:** Aspirin displaces thyroid hormone from thyroid-binding globulin (TBG), **dramatically increasing circulating free T3/T4** and worsening fatal thyrotoxicosis! Use Acetaminophen.



Thyroid Speed Check

When and how should Levothyroxine (Synthroid) be administered?

In the morning on an empty stomach with a full glass of water, 30–60 minutes before breakfast

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ADRENAL DISORDERS: CUSHING'S VS ADDISON'S CRISIS

CUSHING'S VS ADDISON'S: Adrenal cortex hormone dysregulation (Cortisol & Aldosterone). **Cushing's = Excess Cortisol (Moon face, Buffalo hump, Purple striae, Hyperglycemia); Addison's = Deficient Cortisol & Aldosterone (Bronze skin, Hypotension, Hyponatremia, Hyperkalemia)!**

CLINICAL DOMAIN	CUSHING'S SYNDROME (CORTISOL EXCESS)	ADDISON'S DISEASE (ADRENAL INSUFFICIENCY)
Physical Appearance	• Moon Face & Buffalo Hump (truncal obesity). • Thin, fragile skin with Purple Abdominal Striae. • Hirsutism, muscle wasting in thin extremities.	• Hyperpigmentation (Bronze / Sun-tanned skin) on knuckles, knees, elbows, and palmar creases. • Weight loss, muscle weakness, fatigue.
Serum Electrolytes	• Hypokalemia (Low K^+). • Hypernatremia (High Na^+). • Hypocalcemia.	• HYPERKALEMIA (High K^+ / cardiac arrest risk!). • HYPONATREMIA (Low Na^+). • Hypercalcemia.
Blood Glucose & BP	• Hyperglycemia (Steroid diabetes). • Hypertension (Fluid retention).	• Hypoglycemia. • Severe Orthostatic Hypotension & Shock.
Core Medical Therapy	Surgical removal of tumor (Adrenalectomy / Hypophysectomy) or tapering corticosteroid dosage.	Lifelong daily hormone replacement: Hydrocortisone (glucocorticoid) + Fludrocortisone (mineralocorticoid).

ADDISONIAN CRISIS (ACUTE ADRENAL INSUFFICIENCY) EMERGENCY

TRIGGER & CLINICAL SHOCK PRESENTATION

- Triggered by sudden abrupt cessation of steroid therapy, acute infection, or major trauma.
- **Signs:** Profound refractory hypotension, vascular collapse, severe hypoglycemia, severe hyperkalemia, acute abdominal pain.

EMERGENCY RESUSCITATION PROTOCOL

- 1. Immediate rapid IV bolus of **Hydrocortisone Sodium Succinate (100 mg IV)**!
- 2. Aggressive IV fluid resuscitation with **0.9% Normal Saline + 5% Dextrose** (treats hypotension & hypoglycemia).
- 3. Treat hyperkalemia (Calcium gluconate / Insulin).

STEROID ABRUPT WITHDRAWAL TRAP

Patients receiving long-term corticosteroid therapy undergo hypothalamic-pituitary-adrenal (HPA) suppression. **Stopping steroids abruptly precipitates fatal Addisonian Crisis!** Dosages must always be tapered down gradually.

 **Adrenal Speed Check**
What electrolyte pattern classically defines Addison's disease?

Hyponatremia (Low Na^+) coupled with Hyperkalemia (High K^+)

INTEGUMENTARY ASSESSMENT & PEMPHIGUS VULGARIS

 **INTEGUMENTARY ASSESSMENT: Systematic evaluation of skin, hair, and nails. Pemphigus Vulgaris is a potentially fatal autoimmune blistering dermatosis characterized by a Positive Nikolsky's Sign (dislodgement of superficial epidermis by lateral finger pressure)!**

PRIMARY SKIN LESION	MORPHOLOGICAL DEFINITION & SIZE CRITERIA	CLINICAL DISEASE EXEMPLAR
MACULE vs PATCH	Flat, circumscribed, non-palpable color change. • Macule: < 1 cm (freckle, petechiae). • Patch: > 1 cm (vitiligo, mongolian spot).	Vitiligo, measles rash, petechial rash.
PAPULE vs PLAQUE	Elevated, solid, palpable lesion. • Papule: < 1 cm (mole, wart). • Plaque: > 1 cm (broad flat-topped elevation).	Psoriasis plaques, lichen planus, elevated nevus.
VESICLE vs BULLA	Elevated, circumscribed cavity containing clear serous fluid. • Vesicle: < 1 cm (herpes, chickenpox). • Bulla: > 1 cm (blister, pemphigus).	Herpes zoster, second-degree thermal burns, pemphigus.

PEMPHIGUS VULGARIS VS BULLOUS PEMPHIGOID (AIIMS HIGH-YIELD)

PEMPHIGUS VULGARIS (INTRAEPIDERMAL)

- Autoantibodies against desmoglein (desmosomes) → **Acantholysis**.
- Flaccid, fragile, thin-walled blisters that rupture easily, leaving weeping, painful, denuded red lesions.
- Starts in **Oral Cavity** in > 80% of cases!
- **NIKOLSKY'S SIGN IS POSITIVE!**

BULLOUS PEMPHIGOID (SUBEPIDERMAL)

- Autoantibodies against basement membrane hemidesmosomes.
- Tense, firm, thick-walled blisters that do NOT rupture easily.
- Rarely involves oral mucous membranes.
- **NIKOLSKY'S SIGN IS NEGATIVE!**

NIKOLSKY'S SIGN DEFINITION

Firm sliding pressure with a finger over normal-appearing skin causes the epidermis to shear off, slide, and blister: **Positive Nikolsky's Sign!** Pathognomonic for *Pemphigus Vulgaris* and *Toxic Epidermal Necrolysis (TEN / Stevens-Johnson)*.



Skin Assessment Speed Check

What physical skin sign is pathognomonic of Pemphigus Vulgaris?

Positive Nikolsky's Sign (Epidermis separates under lateral rubbing pressure)

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PSORIASIS, MELANOMA (ABCDE) & MEDICATED BATHS

 **PSORIASIS & MELANOMA:** Psoriasis is an autoimmune disease characterized by silvery scales and **Auspitz Sign** (pinpoint bleeding upon scraping scale). Malignant Melanoma is the deadliest skin cancer, screened via the ABCDE mnemonic!

INTEGUMENTARY DISORDER	PATHOPHYSIOLOGICAL FEATURES & CLASSIC LESIONS	THERAPEUTIC MANAGEMENT & NURSING CARE
PSORIASIS VULGARIS	Accelerated epidermal cell turnover (3–4 days instead of 28 days) \$ ightarrow\$ Well-demarcated erythematous plaques covered with silvery white micaceous scales (elbows, knees, scalp).	• Auspitz Sign: Pinpoint bleeding when scale scraped off. • Koebner Phenomenon: Lesions appear at trauma sites. • Topical Corticosteroids, Coal Tar, Phototherapy (PUVA).
MALIGNANT MELANOMA	Abnormal proliferation of melanocytes, leading to invasive growth and potential metastasis.	Wide Surgical Excision

THE ABCDE RULE FOR MALIGNANT MELANOMA SCREENING

ABCDE CRITERIA

- **A = Asymmetry:** One half of the mole does not match the other half.
- **B = Border:** Edges are irregular, notched, scalloped, or blurred.
- **C = Color Variation:** Mottled shades of tan, brown, black, red, white, or blue.
- **D = Diameter:** **> 6 mm** (larger than a standard pencil eraser!).
- **E = Evolving:** Changing in size, shape, color, or new elevation/bleeding.

MEDICATED BATHS (COLLOIDAL OATMEAL)

- Used for generalized pruritus, eczema, and psoriasis.
- Water temperature must be **tepid (lukewarm)**, never hot (hot water triggers intense histamine release and rebound itching!).
- Pat dry gently with towel; do not rub skin vigorously.
- Apply emollients immediately while skin is damp.



Skin Oncology & Psoriasis Speed Check

What physical sign in psoriasis describes pinpoint bleeding when silvery scales are removed?

Auspitz Sign (Pathognomonic indicator of plaque psoriasis)



MSK ASSESSMENT, NEUROVASCULAR CHECKS (6 P's) & DIAGNOSTICS

NEUROVASCULAR ASSESSMENT: Essential clinical monitoring following musculoskeletal trauma, casting, or orthopedic surgery. **The 6 P's: Pain, Paresthesia, Pallor, Pulselessness, Poikilothermia, Paralysis.** Pain with passive stretch is the earliest warning!

ASSESSMENT DOMAIN	CLINICAL EXAMINATION TECHNIQUE & NORMAL BASELINE	ABNORMAL PATHOLOGY & SIGNIFICANCE
PAIN & SENSATION	Assess severity via 0–10 numeric scale. Check light touch and pinprick across dermatomes distal to injury.	Severe pain out of proportion to injury or pain on passive muscle stretch indicates acute ischemia!
PERFUSION & PULSES	Palpate distal peripheral pulses bilaterally (radial, dorsalis pedis, posterior tibial). Assess capillary refill (< 2 seconds).	Weak, thready, or absent pulses; cool pale extremities; delayed refill (> 3 seconds) indicate arterial compromise.
MOTOR FUNCTION	Test active movement of digits (wiggle fingers/toes; dorsiflexion and plantarflexion of ankle).	Inability to move digits indicates nerve impingement, deep tendon laceration, or advanced muscle necrosis.

DIAGNOSTIC IMAGING MODALITIES IN ORTHOPEDICS

PLAIN RADIOGRAPHS & CT SCAN

- **X-Ray:** First-line imaging. Must include **joint above and joint below** the suspected fracture site.
- **CT Scan:** Superior visualization of complex intra-articular fractures, pelvic trauma, and spinal alignment.

MRI & ARTHROSCOPY

- **MRI:** Gold standard for soft tissue injury (ligament tears, meniscal tears, disc herniations, avascular necrosis).
- **Arthroscopy:** Direct endoscopic visualization and surgical repair of joint interior.

PULSELESSNESS IS A LATE SIGN TRAP

Waiting for **pulselessness** before suspecting acute neurovascular compromise or compartment syndrome is a fatal mistake! Pulselessness is an end-stage sign indicating irreversible muscle infarction. Act early!



Neurovascular Speed Check

What is the earliest, most sensitive indicator of acute compartment syndrome?

Disproportionate pain, especially upon passive stretching of distal muscles



FRACTURES: CLASSIFICATION, HEALING & EMERGENCY CARE

 **FRACTURES:** Disruption in the continuity of bone structure. **Open (Compound)** carries massive infection/osteomyelitis risk; **Closed (Simple)** maintains intact skin. Emergency rule: "Splint them where they lie" before moving the client!

FRACTURE PATTERN	STRUCTURAL MORPHOLOGY & MECHANICAL VECTOR	CLINICAL EXEMPLAR / POPULATION
OPEN (COMPOUND)	Bone fragments pierce the skin barrier, creating direct communication with external environment.	High-velocity motor vehicle trauma. High risk of tetanus and acute osteomyelitis!
COMMINUTED	Bone splintered or shattered into 3 or more fragments .	Severe crushing injury or high-energy ballistic trauma. Requires open reduction (ORIF).
GREENSTICK	Incomplete fracture where one side of the bone bends while the opposite cortex fractures.	PEDIATRIC PATIENTS exclusively, due to pliable, soft, flexible bones.
PATHOLOGICAL	Fracture occurring through bone weakened by pre-existing disease during minimal stress.	Osteoporosis, metastatic bone cancer, multiple myeloma, osteomalacia.

THE 4 SEQUENTIAL STAGES OF BONE HEALING

1. HEMATOMA & 2. FIBROCARILAGINOUS CALLUS

- **Stage 1 (Days 1–5):** Hematoma forms around bone ends; inflammatory cytokines trigger granulation.
- **Stage 2 (Weeks 2–3):** Fibroblasts and chondroblasts create a soft cartilage splint (Soft Callus).

3. BONY CALLUS & 4. REMODELING

- **Stage 3 (Weeks 3–12):** Osteoblasts convert soft callus into woven spongy bone (Hard Callus visible on X-ray!).
- **Stage 4 (Months to Years):** Compact lamellar bone remodels along lines of stress (Wolff's Law).

OPEN FRACTURE EMERGENCY TRAP

Never attempt to push protruding bone fragments back beneath the skin! Cover the wound immediately with a **sterile saline-soaked dressing**, administer tetanus toxoid, and initiate broad-spectrum IV antibiotics.

 **Fracture Speed Check**
What unique incomplete fracture type is seen almost exclusively in young children? **Greenstick Fracture (One cortex bends while the other cracks)**

COMPARTMENT SYNDROME & FAT EMBOLISM SYNDROME

🚑 **ORTHOPEDIC EMERGENCIES:** **Compartment Syndrome** (pressure in fascial compartment > 30 mmHg compromises capillary perfusion → Emergent Fasciotomy); **Fat Embolism Syndrome (FES)** occurs 24–72 hours post-long bone fracture!

COMPLICATION	PATHOPHYSIOLOGICAL MECHANISM	CLASSIC CLINICAL PRESENTATION	EMERGENCY ACTION
ACUTE COMPARTMENT SYNDROME	Increased pressure within inelastic fascial compartment (edema/tight cast) exceeds capillary perfusion pressure.	Severe pain out of proportion , pain on passive stretch, tense tight muscle, paresthesia, late pallor.	Bivalve cast, KEEP LIMB AT HEART LEVEL (Do NOT elevate!) , prepare for emergent Fasciotomy!
FAT EMBOLISM SYNDROME (FES)	Fat globules released from yellow marrow of fractured long bones (femur, tibia, pelvis) enter systemic venous circulation.	Triad: 1) Respiratory distress / Hypoxemia ; 2) Neurological changes (confusion) ; 3) Petechial rash on chest & neck!	High-flow O ₂ , supportive ventilation, IV corticosteroids, early fracture immobilization.

COMPARTMENT SYNDROME MANAGEMENT PROTOCOL

THE ELEVATION & ICE TRAPS

- **DO NOT ELEVATE ABOVE HEART LEVEL:** Elevation further lowers arterial perfusion pressure into the compromised compartment!
- **DO NOT APPLY COLD PACKS:** Cold triggers vasoconstriction, accelerating tissue ischemia and muscle necrosis!

FASCIOTOMY PROCEDURE

- Surgical incision through deep fascia to relieve compartment pressure.
- Wound is left open, packed with moist sterile dressings, and closed secondarily once swelling subsides.

FAT EMBOLISM PETECHIAE HALLMARK

The appearance of a **petechial rash across the chest, axillae, and conjunctivae** in a patient with a fractured femur 24 to 72 hours post-injury is pathognomonic for **Fat Embolism Syndrome!** Differentiates FES from pulmonary embolism.



Emergency Speed Check

Why is limb elevation contraindicated in acute compartment syndrome?

Elevation decreases arterial perfusion pressure into the high-pressure compartment

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NUR-08 Batch 4: Units X & XI (Musculoskeletal & Communicable)

CAST CARE, DRYING DYNAMICS & CAST SYNDROME

⚡ **CAST CARE:** Plaster of Paris (takes 24–72 hrs to dry; handle with palms, not fingertips!); Fiberglass (dries in 20–30 mins, water-resistant). **Body spica casts risk Cast Syndrome (Superior Mesenteric Artery Syndrome)!**

CAST MATERIAL	DRYING TIMELINE	HANDLING PRECAUTIONS	PATIENT ADVANTAGES
PLASTER OF PARIS (POP)	24 to 72 hours to dry completely (feels heavy, cool, and damp while wet).	Handle wet cast ONLY WITH OPEN PALMS! Never use fingertips (creates internal indentations causing pressure ulcers!).	Inexpensive, highly moldable; smooth edges. Keep uncovered while drying.
FIBERGLASS (Synthetic)	20 to 30 minutes to achieve full weight-bearing strength.	Warm exothermic reaction during setting; rough edges require moleskin padding (petaling).	Lightweight, rigid, radiolucent on X-rays, water-resistant.

CAST SYNDROME (SUPERIOR MESENTERIC ARTERY SYNDROME)

PATHOPHYSIOLOGY & SIGNS

- Occurs in patients wearing a **Body Jacket / Hip Spica Cast**.
- Cast hyperextension causes Superior Mesenteric Artery (SMA) to compress the **3rd part of the duodenum!**
- **Signs:** Severe abdominal distension, nausea, projectile vomiting, epigastric pain, paralytic ileus.

EMERGENCY NURSING INTERVENTIONS

- 1. Cut or bivalve the cast immediately to relieve abdominal compression.
- 2. Keep patient NPO; insert Nasogastric (NG) tube for decompression.
- 3. Position client in prone or left lateral decubitus position (relieves SMA pressure on duodenum).

CAST DRYING HAIR-DRYER TRAP

Never use an artificial heat lamp or heated blow-dryer to accelerate Plaster of Paris drying! Plaster generates internal exothermic heat; external heat causes **severe underlying skin burns** and uneven drying.



Cast Care Speed Check

Why must a wet plaster cast be handled only with the flat palms of the hands?

Fingertip pressure creates internal plaster indentations leading to pressure ulcers



TRACTION PRINCIPLES: SKIN VS SKELETAL & PIN CARE

 **TRACTION PRINCIPLES:** Application of pulling force to align bone fragments and overcome muscle spasm. **Skin Traction** (Buck's: max 5–7 lbs / 2–3 kg; non-invasive) vs **Skeletal Traction** (Steinmann pin / Kirschner wire: up to 25–40 lbs). Weights must hang freely at all times!

TRACTION TYPE	APPLICATION ROUTE & WEIGHT LIMIT	COMMON CLINICAL INDICATIONS	PRIMARY NURSING RISK
SKIN TRACTION (Buck's / Russell's)	Applied to skin using adhesive tape, boot, or straps. Temporary. • Max Weight: 5 to 7 lbs (2.3–3.2 kg)!	Pre-operative immobilization of Hip Fractures and femoral shaft fractures to relieve muscle spasms.	Skin breakdown, pressure necrosis over Achilles tendon, peroneal nerve compression.
SKELETAL TRACTION	Pins, wires, or tongs surgically inserted directly into bone. • Weight: 15 to 40 lbs (6.8–18 kg).	Fractures of femur, tibia, and cervical spine (Gardner-Wells tongs / Halo). Prolonged use.	Pin site infection & Osteomyelitis! Requires daily sterile chlorhexidine pin care.

GOLDEN RULES OF TRACTION MAINTENANCE

WEIGHT & PULLEY STANDARDS

- **WEIGHTS MUST HANG FREELY AT ALL TIMES!** Never let weights rest on the floor, bed frame, or chair.
- Ropes must stay within the center of pulley grooves.
- Never remove or adjust weights without a specific physician order!

COUNTERTRACTION & ALIGNMENT

- Countertraction is provided by the patient's own body weight.
- Ensure patient's feet do not press against the footboard (elevate foot of bed if needed).
- Maintain straight line of pull along the long axis of the fractured bone.

REMOVING TRACTION WEIGHTS TRAP

Removing or bumping traction weights releases the pulling force, triggering **violent muscle spasms that cause bone fragments to override and shear surrounding neurovascular structures!** Never release weights during bed making.



Traction Speed Check

What is the maximum weight safely applied in adult Buck's skin traction? **5 to 7 pounds (2.3 to 3.2 kg) to prevent skin avulsion and breakdown**



 **CRUTCH WALKING GAITS:** Measure 2–3 fingerbreadths (1–2 inches) between axilla and crutch pad. **Bearing weight on axillary pads causes Radial Nerve Palsy (Crutch Paralysis / Wrist Drop)!** Up with Good, Down with Bad on stairs.

CRUTCH GAIT	SEQUENCE OF MOVEMENT	WEIGHT-BEARING STATUS	CLINICAL INDICATION
4-POINT GAIT	Right crutch → Left foot → Left crutch → Right foot. (3 points on ground constantly).	Partial weight-bearing on both legs .	Bilateral muscle weakness, cerebral palsy, early arthritis. Safest gait.
3-POINT GAIT	Both crutches + affected limb advance together → unaffected strong leg steps forward.	NON-WEIGHT BEARING (NWB) on affected leg.	Fractured leg, post-surgical knee/ankle repair, amputation.
2-POINT GAIT	Opposite crutch and leg move forward simultaneously (Right crutch + Left foot together → Left crutch + Right foot).	Partial weight-bearing on both legs .	Resembles normal human walking; requires good balance.
SWING-THROUGH	Both crutches advance → both legs swing past the crutches simultaneously.	Paraplegia or bilateral lower limb paralysis.	Patients wearing bilateral leg braces; rapid advanced gait.

STAIR CLIMBING & CANE WALKING RULES

STAIRS RULE: "UP WITH GOOD, DOWN WITH BAD"

- **Going UP Stairs:** Step up with **GOOD (unaffected) leg first** → bring crutches and bad leg up.
- **Going DOWN Stairs:** Lower **CRUTCHES and BAD (affected) leg first** → step down with good leg.

CANE WALKING (COAL MNEMONIC)

- **Cane Opposite Affected Leg** (Hold cane on the STRONGER uninjured side!).
- Advance cane simultaneously with the affected weak leg.

CRUTCH AXILLARY RESTING TRAP

Instruct the client to bear all body weight strictly on the **palms of the hands (handgrips)** with elbows flexed ~30°! Resting body weight on the axillary pads compresses the **Radial Nerve**, causing permanent **Crutch Palsy (Wrist Drop)**.



Crutch Walking Speed Check

On which side of the body should a patient hold a single-point walking cane?

On the stronger (unaffected) side of the body (COAL: Cane Opposite Affected Leg)

OSTEOARTHRITIS VS RHEUMATOID ARTHRITIS

OA VS RA: Osteoarthritis (OA: Wear-and-tear cartilage degeneration, asymmetrical, pain worse with activity, Heberden's & Bouchard's nodes); Rheumatoid Arthritis (RA: Autoimmune synovial inflammation, symmetrical, morning stiffness > 1 hr, Swan-neck & Boutonnière deformities)!

CLINICAL PARAMETER	OSTEOARTHRITIS (OA) — DEGENERATIVE	RHEUMATOID ARTHRITIS (RA) — AUTOIMMUNE
Pathophysiology	Progressive mechanical loss and wear of articular hyaline cartilage; formation of subchondral bone spurs (osteophytes).	Autoimmune attack against synovial membrane → Pannus formation , joint destruction, and bony ankylosis.
Joint Involvement	ASYMMETRICAL. Primarily weight-bearing joints (Knees, hips, lumbar spine, cervical spine).	SYMMETRICAL & BILATERAL. Primarily small peripheral joints (PIP, MCP of hands, wrists, feet).
Pain & Stiffness Pattern	Pain increases with weight-bearing activity; relieved by rest . Morning stiffness is brief (< 30 minutes).	Morning stiffness is severe and lasts > 1 hour! Joint pain improves with mild activity.
Classic Hand Deformities	• Heberden's Nodes: Distal interphalangeal (DIP). • Bouchard's Nodes: Proximal interphalangeal (PIP).	• Swan-Neck & Boutonnière deformities. • Ulnar deviation of MCP joints. • Subcutaneous rheumatoid nodules.
Laboratory Tests	Normal ESR and CRP. Negative Rheumatoid Factor (RF).	Positive RF, Anti-CCP antibodies , elevated ESR & CRP.

PHARMACOLOGICAL MANAGEMENT COMPARISON

OSTEOARTHRITIS MANAGEMENT

- First line: **Acetaminophen (Tylenol)** (max 3–4 g/day).
- Topical NSAIDs (Voltaren gel), intra-articular steroid injections.
- Weight loss and low-impact swimming/cycling.

RHEUMATOID ARTHRITIS MANAGEMENT

- First line: **DMARDs (Methotrexate)** to halt progressive joint destruction!
- Biologicals (Anti-TNF / Infliximab, Adalimumab).
- *Methotrexate alert: Teratogenic; monitor CBC/liver enzymes.*

HEBERDEN'S VS BOUCHARD'S NODES TRAP

- **Heberden's Nodes:** Located at the **DIP (Distal Interphalangeal)** joints (*H-D: High/Distal*).
- **Bouchard's Nodes:** Located at the **PIP (Proximal Interphalangeal)** joints (*B-P: Base/Proximal*).

 **Arthritis Speed Check**
What is the duration of morning joint stiffness characteristic of Rheumatoid Arthritis? **Greater than 1 hour (Improves with gentle physical activity)**



METABOLIC BONE DISORDERS & BISPHOSPHONATE SAFETY

⚡ **METABOLIC BONE DISORDERS:** Osteoporosis (Bone resorption exceeds formation → Low bone mineral density / T-score $\leq$ -2.5); Osteomalacia (Inadequate bone mineralization due to Vitamin D deficiency / Rickets in adults); Paget's disease!

BONE DISORDER	PATHOPHYSIOLOGICAL MECHANISM	DIAGNOSTIC HALLMARK	PRIMARY NURSING INTERVENTIONS
OSTEOPOROSIS	Osteoclast bone resorption outpaces osteoblast formation → porous, fragile bones. Postmenopausal estrogen deficiency is #1 cause.	DXA Scan T-score $\leq$ -2.5 SD. Dowager's hump (kyphosis), loss of height, hip/wrist fractures.	Weight-bearing exercise (walking), Calcium + Vitamin D, Bisphosphonates (Alendronate).
OSTEOMALACIA	Inadequate mineralization of organic bone matrix due to severe Vitamin D or Phosphate deficiency.	Low serum 25-OH Vitamin D, elevated Alkaline Phosphatase (ALP). Pseudofractures (Looser's zones).	High-dose Vitamin D3 (Cholecalciferol) supplementation, sunlight exposure, dietary calcium.
PAGET'S DISEASE (Osteitis Deformans)	Disorganized bone remodeling: excessive rapid bone resorption followed by disorganized, weak woven bone deposition.	Markedly elevated Alkaline Phosphatase. Bowed legs, enlarged skull (hat size increases!), cranial nerve compression.	Bisphosphonates (Zoledronic acid), Calcitonin, pain management.

ORAL BISPHOSPHONATE ADMINISTRATION RULES (ALENDRONATE / FOSAMAX)

ESOPHAGITIS PREVENTION PROTOCOL

- Take immediately upon waking in the morning on an **empty stomach with 8 oz (240 mL) of plain water** only.
- **CLIENT MUST REMAIN UPRIGHT (SITTING/STANDING) FOR AT LEAST 30 MINUTES** after taking the pill!

ESOPHAGEAL EROSION TRAP

Lying down within 30 minutes of taking oral Alendronate causes severe acid reflux of the pill into the esophagus, producing **severe ulcerating chemical esophagitis and esophageal perforation!** Never take at bedtime.



Metabolic Bone Speed Check

What DXA scan T-score confirms a clinical diagnosis of Osteoporosis?

T-score $\leq$ -2.5 standard deviations below young adult mean



SPINAL DISORDERS, CAUDA EQUINA & LOGROLLING

 **SPINAL PATHOLOGY: Herniated Nucleus Pulposus (HNP / Slipped Disc)** most common at L4–L5 and L5–S1, causing Sciatica. **Cauda Equina Syndrome** (saddle anesthesia & bowel/bladder incontinence) is an emergent surgical crisis!

SPINAL CONDITION	PATHOPHYSIOLOGICAL MECHANISM & ANATOMICAL SITES	CLINICAL HALLMARKS & NURSING ALERTS
HERNIATED DISC (HNP) & SCIATICA	Rupture of outer annulus fibrosus allows gelatinous nucleus pulposus to protrude posteriorly, compressing exiting spinal nerve roots (L4–S1).	• Sciatica: Radiating shooting pain down posterior thigh to foot. • Positive Straight Leg Raise (Lasègue's Sign). • Paresthesias, motor weakness.
CAUDA EQUINA SYNDROME	Severe bilateral sciatic pain, urinary and/or fecal incontinence, saddle anesthesia, and/or sexual dysfunction.	SURGICAL EMERGENCY! • Saddle anesthesia (perineal sensory loss) + Bowel/bladder incontinence or acute urinary retention. • Severe postural headache.
POTT'S SPINE (Tuberculous Spondylitis)	Hematogenous spread of *Mycobacterium tuberculosis* to vertebral bodies (usually lower thoracic/lumbar). Causes bone necrosis and collapse.	Localized back pain, spinal tenderness, angular kyphotic deformity (Gibbus deformity), cold psoas abscess. Anti-TB therapy (DOTS).

POST-OPERATIVE SPINAL SURGERY: THE LOGROLLING PROTOCOL

LOGROLLING TECHNIQUE

- Used after spinal laminectomy or fusion.
- Requires **2 to 3 healthcare personnel**.
- The patient's head, shoulders, torso, and hips are turned simultaneously as a **single unified rigid block!**
- Keeps spine in absolute neutral anatomical alignment; prevents twisting torque.

POST-SPINE COMPLICATION ASSESSMENT

- **CSF Leak:** Clear halo drainage on surgical dressing; test drainage for glucose! Severe postural headache.
- Neurological checks: Assess toe wiggling, plantarflexion, and urinary retention.



Spine Speed Check

What two clinical signs define the neurosurgical emergency of Cauda Equina Syndrome?

Saddle anesthesia (perineal sensory loss) + Bowel/bladder incontinence or acute urinary retention



TOTAL JOINT ARTHROPLASTY & DISLOCATION PRECAUTIONS

 **TOTAL HIP ARTHROPLASTY (THA):** Surgical joint replacement for severe osteoarthritis or femoral neck fractures. **Posterior hip dislocation precautions: Never flex hip > 90°, Never adduct legs past midline (use Abductor Pillow!), Never internally rotate hip!**

ARTHROPLASTY TYPE	SURGICAL RECONSTRUCTION & PROTOCOL	MAJOR POST-OPERATIVE COMPLICATIONS
TOTAL HIP REPLACEMENT (THA)	Replacement of acetabulum (polyethylene cup) and femoral head (titanium stem with ceramic/metal ball).	• Prosthetic Dislocation (shortened, internally rotated limb!). • DVT / Pulmonary Embolism. • Periprosthetic joint infection.
TOTAL KNEE REPLACEMENT (TKA)	Resurfacing of distal femur, proximal tibia, and patella with metal and polyethylene components.	• Joint stiffness and arthrofibrosis. • Deep vein thrombosis. • Use Continuous Passive Motion (CPM) machine!

POST-OP TOTAL HIP DISLOCATION PRECAUTIONS (POSTERIOR APPROACH)

THE "DO NOT" RULES

- **DO NOT flex hip past 90 degrees:** Avoid low chairs, low toilet seats (use elevated raised toilet seats!).
- **DO NOT cross legs** at knees or ankles (Adduction).
- **DO NOT bend forward** to tie shoes or pick items off floor.
- DO NOT internally rotate toes/knee inward.

THE "DO" NURSING PROTOCOLS

- Place a wedge-shaped **Abductor Pillow** between patient's legs while in bed and turning.
- Keep knees apart; use long-handled shoehorns and reachers.
- Early ambulation on post-op day 1 with physical therapy.

HIP PROSTHETIC DISLOCATION SIGNS

If a post-THA patient suddenly screams in acute groin pain and the surgical leg appears **shortened and internally rotated with inability to move**: The prosthesis has dislocated! Notify orthopedic surgeon immediately.



Arthroplasty Speed Check

What physical positioning device is placed between patient legs post-hip replacement?

Abductor Wedge Pillow (Prevents adduction past midline and prosthetic dislocation)



INFECTION TRANSMISSION & ISOLATION PRECAUTIONS

 **ISOLATION PRECAUTIONS: Standard Precautions applied to ALL patients.** Transmission-Based: Contact (Gloves + Gown); Droplet (Surgical Mask within 3–6 feet); Airborne (N95 Respirator + Negative Pressure Room with HEPA filter)!

ISOLATION TIER	TRANSMISSION MECHANISM & PARTICLE SIZE	REQUIRED PPE & ROOM SETUP	CLINICAL DISEASE EXEMPLARS
CONTACT PRECAUTIONS	Direct physical contact with patient or indirect contact with contaminated fomites.	Gloves + Gown before entering room. Dedicated patient equipment (BP cuff, stethoscope).	• MRSA & VRE • <i>Clostridioides difficile</i> (C. diff) • Scabies, Norovirus, RSV.
DROPLET PRECAUTIONS	Large moist droplets ($> 5\ \mu\text{m}$) expelled by coughing/sneezing; travel short distances (3 to 6 feet).	Surgical Mask upon entering room. Private room (or cohort with same organism).	• Meningococcal Meningitis • Pertussis (Whooping cough) • Influenza, Mumps, Rubella.
AIRBORNE PRECAUTIONS	Tiny droplet nuclei ($\leq 5\ \mu\text{m}$) remaining suspended in air currents over long distances and time.	N95 Respirator (or PAPR) + Negative Pressure Room (AIIR) with 6–12 air exchanges/hr.	• Tuberculosis (Pulmonary TB) • Measles (Rubeola) • Chickenpox (Varicella) • Disseminated Herpes Zoster.

CLOSTRIDIoidES DIFFICILE (C. DIFF) HAND HYGIENE MANDATE

SOAP & WATER RULE

- **ALCOHOL-BASED HAND RUBS DO NOT KILL C. DIFF SPORES!**
- Nurses must wash hands vigorously with **Antimicrobial Soap and Running Water** to physically wash spores down the sink!
- Clean room surfaces exclusively with chlorine bleach wipes.

⚡ AIRBORNE MNEMONIC: "MTV":

Remember the 3 classic airborne infections requiring N95 mask + negative pressure room:

- **M** = Measles (Rubeola)
- **T** = Tuberculosis
- **V** = Varicella (Chickenpox / Disseminated Shingles).



Isolation Speed Check

Why are alcohol-based hand sanitizers ineffective after caring for a C. difficile patient?

Alcohol cannot penetrate or destroy bacterial endospores; soap and water is required

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NUR-08 Batch 4: Units X & XI (Musculoskeletal & Communicable)

TUBERCULOSIS (TB): MANTOUX, DOTS & ADVERSE EFFECTS

 **TUBERCULOSIS (TB):** Infection caused by *Mycobacterium tuberculosis* (acid-fast bacillus). **Mantoux TST read 48–72 hours later (measure induration, not erythema!).** First-line DOTS: RIPE (Rifampicin, Isoniazid, Pyrazinamide, Ethambutol)!

FIRST-LINE ANTI-TB DRUG	MECHANISM OF ACTION & ADMINISTRATION RULE	PATHOGNOMONIC ADVERSE EFFECT & ALERT
R = RIFAMPICIN (RIF)	Inhibits bacterial RNA synthesis (bactericidal). Take on empty stomach.	Harmless red-orange discoloration of urine, tears, saliva, and sweat! Stains soft contact lenses. Hepatotoxicity.
I = ISONIAZID (INH)	Inhibits mycolic acid synthesis in mycobacterial cell wall (primary drug).	• Peripheral Neuropathy (prevented by co-prescribing Vitamin B6 / Pyridoxine!). • Severe Hepatotoxicity (monitor AST/ALT).
P = PYRAZINAMIDE (PZA)	Active against intracellular bacilli in acidic macrophages.	Hyperuricemia & Gouty Arthritis! Hepatotoxicity.
E = ETHAMBUTOL (EMB)	Inhibits arabinosyl transferase (bacteriostatic). Safe in liver disease.	Optic Neuritis: Loss of Red-Green Color Discrimination and decreased visual acuity! Monthly vision checks.

MANTOUX TUBERCULIN SKIN TEST (TST / PPD) INTERPRETATION**ADMINISTRATION & TIMING**

- Inject 0.1 mL of PPD intradermally on volar forearm (creates a 6–10 mm wheal).
- Read test result strictly between **48 to 72 hours**.
- **MEASURE INDURATION (HARD RAISED PALPABLE AREA), NEVER ERYTHEMA (REDNESS)!**

POSITIVE INDURATION CUTOFFS

- **≥ 5 mm:** Positive in HIV/immunocompromised, recent TB contacts, organ transplants.
- **≥ 10 mm:** Positive in healthcare workers, high-risk immigrants, IV drug users.
- **≥ 15 mm:** Positive in general population with zero known risk factors.

ETHAMBUTOL COLOR VISION TRAP

A client receiving Anti-TB therapy reports difficulty distinguishing between red and green traffic lights. Ethambutol must be **discontinued immediately** to prevent permanent blindness from toxic optic neuritis!

**Tuberculosis Speed Check**

Which vitamin is routinely co-administered with Isoniazid (INH) to prevent peripheral neuropathy? **Vitamin B6 (Pyridoxine)**

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VECTOR-BORNE DISEASES: MALARIA & DENGUE (DHF)

VECTOR-BORNE INFECTIONS: Malaria transmitted by female *Anopheles* mosquito (night-biter; *P. falciparum* causes cerebral malaria); Dengue transmitted by *Aedes aegypti* (day-biter; "tiger mosquito"). **Dengue Hemorrhagic Fever warning: Thrombocytopenia + Plasma leakage!**

VECTOR DISEASE	PATHOGEN & MOSQUITO VECTOR	CLASSIC CLINICAL PRESENTATION	EMERGENCY NURSING PRIORITIES
MALARIA	*Plasmodium* species (*P. vivax, P. falciparum, P. malariae, P. ovale*). Vector: Female Anopheles mosquito (night-biter).	Classic Paroxysm Triad: 1) Cold Stage (intense shivering); 2) Hot Stage (high fever > 40°C); 3) Sweating Stage (drenching diaphoresis).	Artemisinin-based combination therapy (ACT), IV Artesunate for severe/cerebral malaria, monitor for blackwater fever (hemoglobinuria).
DENGUE FEVER (<i>"Breakbone Fever"</i>)	Dengue Flavivirus (Serotypes 1–4). Vector: Aedes aegypti (day-biter / tiger mosquito).	High fever, severe retro-orbital eye pain, severe joint/muscle pain (breakbone), maculopapular rash. Positive Tourniquet Test!	Judicious IV fluid hydration (Normal Saline/Ringer's). AVOID ASPIRIN / NSAIDs (Fatal hemorrhage risk!). Paracetamol only.

DENGUE WARNING SIGNS & DENGUE SHOCK SYNDROME (DSS)

WHO WARNING SIGNS OF SEVERE DENGUE

- Appears when fever subsides (Days 3 to 7 / Critical Phase!).
- Severe abdominal pain and persistent vomiting.
- Mucosal bleeding (epistaxis, gum bleeding, hematemesis).
- Rapid platelet drop coupled with **hematocrit spike > 20%** (indicates severe plasma leakage!).

TOURNIQUET TEST (RUMPEL-LEEDE)

- Inflate BP cuff midway between systolic and diastolic for 5 minutes.
- Release cuff, wait 2 minutes, count petechiae inside a 1-inch square.
- **Positive Test: ≥ 10 to 20 petechiae per square inch** (indicates capillary fragility!).

DENGUE ASPIRIN CONTRAINDICATION TRAP

Administering Aspirin or Ibuprofen to relieve fever in a Dengue patient severely destroys platelet function and accelerates plasma leakage, triggering fatal **Dengue Hemorrhagic Shock!** Administer Acetaminophen (Paracetamol) strictly.



Vector-Borne Speed Check

Which mosquito species transmits Dengue fever and when does it predominantly bite?

Aedes aegypti (Day-biting mosquito / tiger mosquito)



ENTERIC INFECTIONS: TYPHOID, CHOLERA & HEPATITIS

⚡ **ENTERIC INFECTIONS:** Fecal-oral transmission. **Typhoid Fever** (**Salmonella typhi**: Step-ladder fever, relative bradycardia, Rose spots, Widal test); **Cholera** causes massive "rice-water" diarrhea; **Hepatitis A & E** are fecal-oral!

ENTERIC INFECTION	ETIOLOGY & TRANSMISSION	PATHOGNOMONIC SIGNS	DIAGNOSTIC / NURSING CARE
TYPHOID FEVER (Enteric Fever)	* <i>Salmonella enterica</i> * serotype Typhi. Ingestion of contaminated food/water → Peyer's patch ulceration.	• Step-ladder pyrexia. • Relative Bradycardia (Faget's sign). • Rose spots on trunk. • Pea-soup diarrhea or constipation.	Blood culture (Week 1 / BASU mnemonic), Widal test (Week 2). Complication: Ileal perforation in week 3! Ceftriaxone.
CHOLERA	* <i>Vibrio cholerae</i> * enterotoxin causes massive hypersecretion of water and electrolytes into small bowel lumen.	Painless, profuse, voluminous "Rice-Water Stools" with fishy odor (up to 1 L/hr!). Rapid hypovolemic shock!	Aggressive immediate oral and IV rehydration (WHO ORS / Ringer's Lactate), Doxycycline/ Azithromycin.

VIRAL HEPATITIS: TRANSMISSION ROUTES & ANTIGEN SEROLOGY

HEPATITIS VIRUS	TRANSMISSION ROUTE	CHRONIC INFECTION & CARRIER STATE?	VACCINE AVAILABLE?
HEPATITIS A (HAV)	Fecal–Oral (Food/water)	NO (Acute self-limiting only; zero carrier).	YES (HAV vaccine available).
HEPATITIS B (HBV)	Blood, Body Fluids, Sexual, Perinatal	YES (Chronic cirrhosis & HCC cancer!)	YES (HBsAg recombinant vaccine).
HEPATITIS C (HCV)	Blood transfusion, IV drug use	YES (> 80% chronicity!)	NO VACCINE AVAILABLE!
HEPATITIS D (HDV)	Blood & Body fluids (Requires HBV!).	YES (Superinfection with HBV accelerates failure).	Prevented by Hepatitis B vaccine!
HEPATITIS E (HEV)	Fecal–Oral (Contaminated water)	NO (Except: High fatal fulminant mortality in pregnant women!).	No widespread vaccine.



Enteric Speed Check

Which viral hepatitis strain has the highest maternal mortality rate due to fulminant hepatic failure?

Hepatitis E Virus (HEV; up to 20% mortality in pregnant women during 3rd trimester)



DIPHTHERIA, PERTUSSIS, TETANUS & POLIOMYELITIS

 **VACCINE-PREVENTABLE INFECTIONS:** Diphtheria (**Corynebacterium**: grayish-white pseudomembrane); Pertussis (paroxysmal whooping cough); Tetanus (**Clostridium tetani**: tetanospasmin neurotoxin → Trismus/lockjaw, Opisthotonus, risus sardonicus)!

INFECTION	MICROBIAL ETIOLOGY & TOXIN	CLASSIC PATHOGNOMONIC FEATURES	EMERGENCY NURSING ROLE
DIPHTHERIA	* <i>Corynebacterium diphtheriae</i> * (exotoxin inhibits cellular protein synthesis).	Adherent, thick, Grayish-White Pseudomembrane covering tonsils/pharynx. "Bull Neck" cervical lymphadenopathy.	Diphtheria Antitoxin (DAT) IV + Erythromycin. Keep emergency tracheostomy tray at bedside!
PERTUSSIS (Whooping Cough)	* <i>Bordetella pertussis</i> * (pertussis toxin causes respiratory ciliary paralysis).	Paroxysmal coughing attacks ending in high-pitched inspiratory "Whoop" followed by post-tussive vomiting.	Droplet precautions, humidified oxygen, suctioning, Azithromycin/Erythromycin.
TETANUS ("Lockjaw")	* <i>Clostridium tetani</i> * spores in soil enter dirty puncture wounds → Tetanospasmin blocks inhibitory glycine/GABA!	• Trismus (Lockjaw) & Risus Sardonicus. • Opisthotonus (severe arching spasm of back). • Reflex spasms triggered by noise/light!	DARK, QUIET, SOUNDPROOF ROOM! Tetanus Immunoglobulin (TIG), Diazepam muscle relaxants, wound debridement.

CLINICAL NURSING ALERTS: DIPHTHERIA & TETANUS

DIPHTHERIA MEMBRANE FORCEFUL REMOVAL TRAP

NEVER attempt to forcefully peel or scrape away the grayish pseudomembrane in diphtheria! The membrane is firmly adherent to underlying mucosal capillaries; peeling causes profuse, uncontrolled airway hemorrhage and fatal asphyxiation!

⚡ TETANUS SENSORY ISOLATION MANDATE:

Tetanus toxin causes uninhibited motor nerve firing.

→ The slightest sensory stimulus (slamming door, turning on bright light, loud talking) triggers **violent, agonizing, whole-body tetanic muscle spasms and respiratory arrest!** Nurse client in absolute darkness and quiet.



Vaccine-Preventable Speed Check

What is the severe generalized backward arching of the spine and neck seen in tetanus termed?

Opisthotonus (Severe tetanic muscular hyperextension spasm)



VIRAL EXANTHEMS: MEASLES, VARICELLA & MUMPS

⚡ **VIRAL EXANTHEMS:** Measles (Rubeola: Koplik spots on buccal mucosa); Chickenpox (Varicella: Pleomorphic "dewdrops on a rose petal" rash); Mumps (Parotitis). **Airborne precautions for Measles & Varicella!**

EXANTHEM	VIRAL PATHOGEN & TRANSMISSION	PATHOGNOMONIC RASH & CLINICAL CLUES	MAJOR COMPLICATIONS
MEASLES (Rubeola)	Measles Paramyxovirus. • Airborne Transmission! (Highly contagious).	• 3 C's: Cough, Coryza, Conjunctivitis. • KOPLIK'S SPOTS: Tiny bluish-white spots on red base opposite 2nd lower molars! • Maculopapular rash spreading hairline → downward.	Otitis media, bronchopneumonia, SSPE (Subacute Sclerosing Panencephalitis years later!).
CHICKENPOX (Varicella)	Varicella-Zoster Virus (VZV / HHV-3). • Airborne & Contact Transmission!	PLEOMORPHIC RASH: All stages present simultaneously (macules, papules, clear vesicles resembling " Dewdrops on a rose petal ", and crusts/scabs!).	Secondary bacterial staph infection, cerebellar ataxia, pneumonia. Contagious until all lesions crust!
MUMPS (Epidemic Parotitis)	Mumps Rubulavirus. • Droplet Transmission.	Painful, tender swelling and enlargement of the Parotid Salivary Gland (Parotitis) (unilateral or bilateral), earache on chewing.	Orchitis (testicular inflammation in post-pubertal males → sterility risk!), meningitis, pancreatitis.

KOPLIK'S SPOTS (MEASLES DIAGNOSTIC CLUE)

- Appear **24 to 48 hours BEFORE** the skin rash erupts.
- Located on the oral buccal mucosa opposite the lower 2nd molars.
- Pathognomonic hallmark allowing instant clinical diagnosis!

ASPIRIN IN VIRAL INFECTIONS (REYE'S SYNDROME)

Giving Aspirin to children or adolescents with Chickenpox, Measles, or Influenza is **FATAL!** Triggers **Reye's Syndrome:** acute hepatic fatty encephalopathy and cerebral edema! Use Acetaminophen.

🎯 Exanthems Speed Check

What oral mucosal lesion is pathognomonic of early Measles (Rubeola)?

Koplik's Spots (Tiny bluish-white granular spots opposite lower molars)

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NUR-08 Batch 4: Units X & XI (Musculoskeletal & Communicable)

LEPROSY (HANSEN'S) & GAS GANGRENE EMERGENCY

LEPROSY & GANGRENE: Leprosy (*Mycobacterium leprae*: hypopigmented anesthetic skin patches, thickened peripheral nerves); Gas Gangrene (*Clostridium perfringens*: subcutaneous crepitus, foul sweet drainage) is a limb- and life-threatening surgical emergency!

BACTERIAL INFECTION	PATHOGEN & INCUBATION PERIOD	PATHOGNOMONIC CLINICAL PRESENTATION	STANDARD TREATMENT
LEPROSY (Hansen's Disease)	*Mycobacterium leprae* (acid-fast bacillus). Prolonged incubation period (3 to 5 years!). Transmitted via droplet contact with nasal mucosa.	• Hypopigmented skin patch with loss of sensation (touch/pain/temp). • Thickened, palpable peripheral nerves (ulnar, greater auricular). • Leonine Facies in lepromatous.	WHO Multi-Drug Therapy (MDT): Rifampicin + Dapsone + Clofazimine . (Disability prevention & footwear education).
GAS GANGRENE (Clostridial Myonecrosis)	*Clostridium perfringens* (anaerobic spore-forming bacillus). Enters deep traumatic, devitalized, soil-contaminated puncture wounds.	• Sudden severe wound pain. • Palpable subcutaneous CREPITUS (gas bubbles under skin!). • Thin, foul, sweet watery brownish discharge.	EMERGENCY EXTENSIVE SURGICAL DEBRIDEMENT / Amputation + High-dose IV Penicillin G + Hyperbaric O2!

CLINICAL SIGNS: GAS GANGRENE VS LEPROSY REACTION

PALPABLE CREPITUS IN GAS GANGRENE

- Palpating the skin around a traumatized wound produces a **crackling, bubbling sensation (like bubble wrap)**.
- Caused by gas produced by anaerobic carbohydrate fermentation of *C. perfringens*!
- ✓ **Action:** Prepare for emergency operating room debridement immediately.

LEPRA REACTIONS (TYPE 1 & TYPE 2)

- **Type 1 Reaction:** Cell-mediated immune reversal reaction; severe acute nerve inflammation (neuritis) → risk of permanent claw hand!
- **Type 2 Reaction (ENL):** Erythema Nodosum Leprosum; painful erythematous nodules, fever, joint pain. Treated with Thalidomide / Steroids.

 Leprosy & Gangrene Speed Check

What physical palpation finding is diagnostic of acute gas gangrene around a traumatic wound? **Subcutaneous Crepitus (Crackling sensation due to gas bubbles in tissue)**

BARRIER NURSING, NEEDLE-STICK PROTOCOL (PEP) & BMW

SAFETY PROTOCOLS: Needle-Stick Injury Protocol: Wash with soap & water immediately (Never squeeze/milk!); initiate Post-Exposure Prophylaxis (PEP) within 2 to 72 hours. Biomedical Waste (BMW) Color Coding: Red (plastic), Yellow (anatomical/cotton), White (sharps), Blue (glassware)!

BMW WASTE BIN	AUTHORIZED WASTE CATEGORIES INCLUDED	TREATMENT & DISPOSAL METHOD	COMMON HOSPITAL ITEMS
YELLOW BAG	Human anatomical waste, animal tissues, soiled cotton, gauze dressings, expired medications, cytotoxic drugs.	Incineration or Plasma Pyrolysis / Deep burial.	Placenta, amputated limbs, blood-soaked swabs, soiled dressings.
RED BAG	Contaminated recyclable plastics only. (Autoclaved / shredded).	Autoclaving / Microwaving + Shredding → Recycling.	IV bottles, IV tubing, Foley catheters, urine bags, disposable syringes (without needles!).
WHITE CONTAINER (Puncture-Proof)	Metallic sharp objects (needles, scalpels, blades, contaminated needles with fixed syringes).	Autoclaving + Dry Heat Sterilization → Shredding / Encapsulation in metal sharp pit.	Needles, lancets, scalpels, suture needles, trocars.
BLUE BOX / CARDBOARD	Glassware and metallic body implants.	Disinfection (Soaking in sodium hypochlorite) + Autoclaving / Recycling .	Broken ampoules, vials, glass medicine bottles, orthopedic metal screws/plates.

NEEDLE-STICK INJURY (NSI) & POST-EXPOSURE PROPHYLAXIS (PEP) PROTOCOL**IMMEDIATE FIRST AID ACTIONS**

1. **WASH IMMEDIATELY WITH SOAP AND RUNNING WATER!**
2. **DO NOT SQUEEZE, PRESS, OR MILK THE WOUND!** (Squeezing forces pathogens deeper into tissue).
3. **DO NOT USE ANTISEPTICS** like bleach or alcohol (causes tissue irritation).
4. If splash to eyes/mucosa: Flush with copious sterile saline/water for 15 minutes.

PEP TIMELINE & REGIMEN

- Report incident immediately to infection control officer.
- **Initiate HIV PEP ideally within 2 HOURS (Maximum 72 hours window!).**
- Standard 28-day antiretroviral course (Tenofovir + Lamivudine + Dolutegravir / TLD regimen).

NEEDLE RECAPPING PROHIBITION TRAP

NEVER RECAP USED NEEDLES WITH TWO HANDS! Recapping is the #1 cause of occupational needle-stick injuries in hospital wards. If recapping is absolutely unavoidable, use the single-handed "Scoop" technique, or drop immediately into a puncture-proof white sharp box!

 **Safety & BMW Speed Check**
In which biomedical waste container category must used disposable syringe plastic barrels (without needle) be discarded?

Red Bag (Contaminated recyclable plastic waste)

